Week 4 · Heading Control

USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering, Chungnam National University | revised 2026-09-21

ℹ Recordings of this semester's own classes

Every session of this course is recorded as it is given and posted to the playlist below. The written material here is the reference; the recording is the lecture, at the pace it was actually delivered. Use it to review a derivation, or to catch up on a missed week before the laboratory.

	Where
[충남대] 2026 센서신호처리 — this course, recorded live, 2026	playlist

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	**Control Engineering** (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	**Control System Design** (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	**Advanced Control Engineering** (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	**Sensor Signal Processing and Fusion** (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	**Capstone Design** (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

✎ Getting the course files, and keeping them current (Windows)

The notes, models and scripts are kept in one Git repository that is updated through the semester. Clone it once; before every class, pull. A pull downloads only what has changed since the last one.

When	Where to run it	Command
once	PowerShell — installs Git for Windows	`winget install --id Git.Git -e`
once	the folder that will hold the course, e.g. `Documents`	`git clone https://github.com/wkyouncnu/Sensor-Signal-Processing-and-Fusion.git`
before every class	inside the cloned folder `Sensor-Signal-Processing-and-Fusion`	`git pull`

- `git pull` prints `Already up to date.` when nothing has changed, and otherwise lists the files it updated.
- The repository is private. When Git asks, sign in with the GitHub account the instructor has given access.
- Experiment on copies, not on the cloned files: copy a week's `WXX_simulink` folder elsewhere first, and a pull can never collide with local edits. If it already has, `git stash`, then `git pull`, then `git stash pop` sets the edits aside, updates, and puts them back.
- The MSS toolbox is not part of the repository. The weeks that simulate the Otter need it at `Tools\MSS` inside the cloned folder.

- **Course**: USV Guidance, Navigation and Control (Graduate)
- **Department**: Autonomous Vehicle System Engineering, Chungnam National University
- **This week**: ① the Week 2 controller around the Otter's heading, tuned from the Scope with no model of the vessel ② why P alone reaches the command here and D damps again — the opposite of Week 3 ③ the crab angle: heading is not course ④ the moment limit, the back-calculation gain, and the wrap at $\pm 180^\circ$

★ Prerequisites from the previous week

- From Weeks 2 and 3: the five time-domain metrics, what P, I and D do, the filtered derivative, back-calculation anti-windup and the tuning order. Nothing new is added to the controller this week.
- From Week 3 §3-3: the same derivative term that damps a position loop made the speed loop worse. A term's effect is read from the axis, not assumed.
- From Week 1: the two propellers sit $y_{\text{pont}} = 0.395$ m either side of the centreline, so a difference in their thrust turns the vessel.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. Characterise the yaw axis from the outside with one step of yaw moment, and explain from the result why P alone leaves no heading error.
2. Predict, from Weeks 2 and 3 and one fact about the axis, how P, I and D each change the heading response, and confirm it on the Scope.
3. Distinguish heading from course, and compute the crab angle from the body velocities.
4. Remove the heading error left by a weak propeller with the integral, and choose the back-calculation gain from a measured trade-off.
5. Explain why a heading error must be wrapped into $(-180^\circ,\ 180^\circ]$, and measure the cost of not wrapping it.

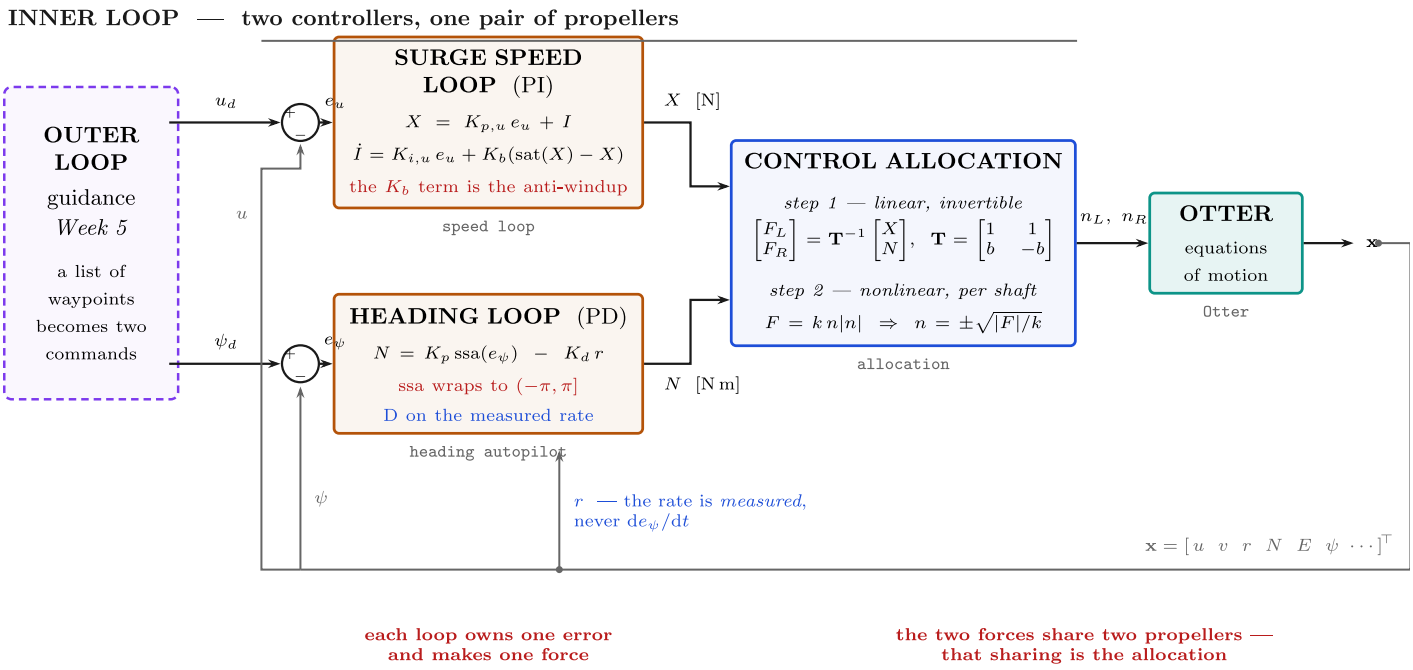
6. Write the complete heading law in one line — ssa on the error, the rate term on the measurement, the clamp on the output — and justify each of the three from a measurement made this week.
7. Follow the commanded moment through the allocator of §3-6 to a pair of shaft speeds, and explain why the moment limit puts one propeller exactly on its own limit.
8. Run the speed loop of Week 3 and the heading loop of Week 4 on one hull, and state where the two controllers interact and where they do not.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b, with Simulink
MSS	`Tools/MSS`, found by `W04_0_setup` through `mss_path`
Course folder	`lectures/W04_simulink`
Models	one per example — `W04_C_open_loop`, `W04_D_P`, `W04_E_PD`, `W04_F_PID`, `W04_G_wrap`, `W04_H_tuning` — all generated by `W04_1_build_heading`, never edited by hand
Time	lecture 40 min, laboratory 1 h 20 min

Where this week sits

The vessel has two controllers and one pair of propellers. Week 3 built the upper row of the diagram below and left the lower one empty; this week fills it in. Everything to the right of the two controllers — the allocator and the hull — is already finished and is not touched.



In the diagram	What it is	Where it was built
upper row, `speed loop`	a PI on the speed error, with anti-windup, producing the surge force X	Week 3 §3-3, §3-4
lower row, `heading autopilot`	**a PD on the heading error, producing the yaw moment N**	**this week**
`allocation`	turns (X, N) into the two shaft speeds, in two steps	Week 3 §3-6
`Otter`	the equations of motion	Week 1
dashed box, left	where ψ_d and u_d come from once a path exists	Week 5
feedback bus, bottom	the state vector; the heading loop reads ψ **and** r, and the reason it reads r separately is §4-3d	

- The two controllers never exchange a signal. They meet only in the allocator, and §4-7 is where that meeting is worked out with this week's numbers.
- Read the lower row's contents as the destination of the week: by §4-7 every symbol in it will have been forced by a measurement.

Part 1 · Principle and experiment, section by section

Every section states what is observed, says what follows from it, and then runs **the experiment that measures it**, on its own model. The numbers quoted in the text are printed by the run a few lines below it.

Part of a section	What it holds
What is observed	the effect itself, with the numbers it produces
What follows from it	numbered lines, one step each, resting on Weeks 2 and 3 and on one measurement of this axis
Experiment N-x	the model, the commands, the output actually obtained, the figure, and a table that reads every feature of the figure back to **the numbered line that predicts it**

4-0. Setting up (10 min)

```
cd lectures/W04_simulink
W04_0_setup
W04_1_build_heading
```

Expected output:

```
  W04_0_setup
    thrust   X_ff = 60 N forward, yaw moment |N| <= 70.85 N m
    gains    Kp = 300   Ki = 20   Kd = 100   Nf = 20   Kb = 0.1
    command  psi_d = 0 -> 10 deg at t = 5 s

  built  W04_C_open_loop.slx  (overlapping lines: 0)
  built  W04_D_P.slx          (overlapping lines: 0)
  built  W04_E_PD.slx         (overlapping lines: 0)
  built  W04_F_PID.slx        (overlapping lines: 0)
  built  W04_G_wrap.slx       (overlapping lines: 0)
  built  W04_H_tuning.slx     (overlapping lines: 0)
  built  W04_I_rate.slx       (overlapping lines: 0)
```

- `W04_0_setup.m` is the only file edited by hand. If it stops in `mss_path`, the MSS toolbox is not at `Tools\MSS`.
- The first line of `W04_0_setup` is `clear`. Run it again whenever a Scope does not match these notes.

4-1. The same controller, a new output

This section answers: what changes when the controller of Weeks 2 and 3 steers the vessel instead of speeding it up?

The controller does not change. The error, the three terms, the derivative filter, the limit and the back-calculation path are the blocks of Week 3, in the same places. Three things around them are new.

New block	What it does	Why
`deg to rad`	converts the command from degrees to radians	people command headings in degrees; the vessel model works in radians
`ssa`	wraps the error into $(-\pi,\ \pi]$: $\mathrm{ssa}(e) = \mathrm{atan2}(\sin e,\ \cos e)$	-170° is 20° past 170°, not 340° back; §4-6
`allocation`	a MATLAB Function block, commented line by line: turns the demanded yaw moment N into two propeller thrusts, then two shaft speeds	the Otter has no rudder; it turns by pushing one side harder

The yaw moment comes from a thrust difference. Both propellers push forward to give the surge force X_{ff}; a yaw moment is added by pushing the port propeller harder and the starboard one less. Each propeller sits y_{pont} from the centreline, so a thrust difference $T_1 - T_2$ gives a moment $y_{\text{pont}}(T_1 - T_2)$, and the two thrusts are

$$T_1 = \frac{X_{ff}}{2} + \frac{N}{2\,y_{\text{pont}}}\ \text{(port)}, \qquad T_2 = \frac{X_{ff}}{2} - \frac{N}{2\,y_{\text{pont}}}\ \text{(starboard)}$$

Symbol	Quantity	Value · source
X_{ff}	surge force from both propellers together	60 N, `W04_0_setup.m`
N	yaw moment demanded by the controller	N·m
y_{pont}	propeller arm, half the distance between the pontoons	0.395 m, `otter_config`
$N_{\max}$	largest yaw moment available while X_{ff} is delivered	70.85 N·m: the port propeller reaches its 119.68 N maximum at $N = 2\,y_{\text{pont}}\,(119.68 - 30)$

- Each thrust is then turned into a shaft speed exactly as in Week 3, $n = \mathrm{sign}(T)\sqrt{|T|/k}$.
- The `moment limit` block holds N inside ± 70.85 N·m. It plays the role the thrust limit played in Week 3, and it causes the windup of §4-5.

Experiment 4-1 · The Week 3 controller, steering instead of speeding up (10 min)

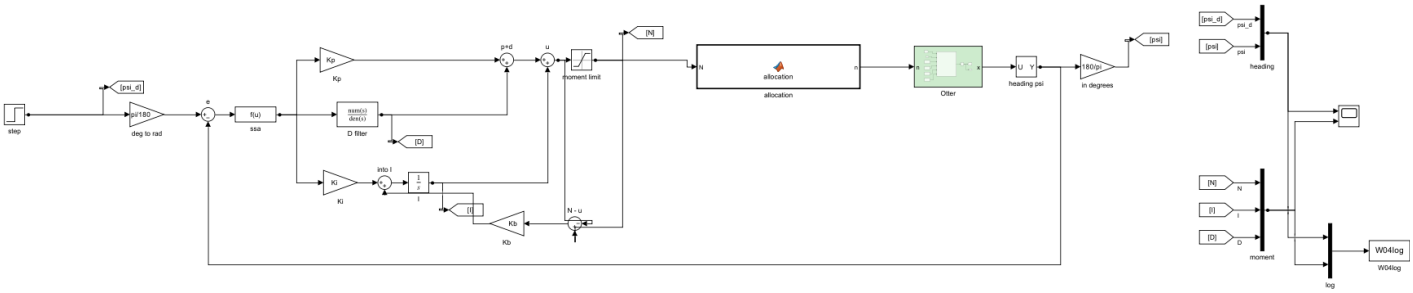
What it measures. Nothing yet: the canvas is matched to the three new blocks and to the thrust split of the formula above, so that each is recognised when it matters.

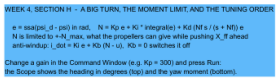
The canvas. `W04_H_tuning` — the fullest model of the week, read here and measured in Experiment 4-5. Everything between `e` and `u` is the Week 3 canvas unchanged.

Opening and running.

```
W04_0_setup
open_system('W04_H_tuning')        % Run — 위 칸 선수각, 아래 칸 모멘트 / heading above, moment below
```

In the law	On the canvas	Which changed since Week 3
the command in degrees	`step` → `deg to rad`	new: people command headings in degrees
$e = \mathrm{ssa}(\psi_d - \psi)$	`e` → `ssa`	new: the wrap of §4-6
$K_p e + K_i \int e - K_d r$	`Kp` , `Ki` → `I` , `D filter`	the derivative now acts on the measured rate (§4-3 line 4)
the actuator's limit	`moment limit`	the quantity: ± 70.85 N·m instead of a force
T_1, T_2 and two shaft speeds	`allocation`	new: a moment out of two propellers





In the figure	Meaning
`step` → `deg to rad` → `e` → `ssa`	the command in degrees, the error in radians, wrapped
`Kp` , `D filter` , `Ki` → `I` , `p+d` , `u`	the PID of Weeks 2 and 3, block for block
`moment limit` , `N - u` , `Kb`	the moment limit and the back-calculation path of Week 3
`allocation`	T_1 and T_2 of §4-1, the port propeller's efficiency `port_eff` (1 unless Experiment 4-3c lowers it), and the two shaft speeds; double-click it to read the commented code
`Otter` → `heading psi` → `in degrees`	the vessel, and the heading back in degrees for the Scope
`heading` , `moment` → Scope	top: ψ_d and ψ; bottom: N, I and D

> ✎ **In class**
>
> - **Purpose** — show that the controller is still the Week 2 controller; the new parts are the degree conversion, the wrap, and the thrust split that makes a moment out of two propellers.
> - **Point to** — open the `allocation` block behind the `moment limit` : one moment in, two thrusts out, one larger and one smaller, each turned into a shaft speed.
> - **Ask** — "Why is there a `moment limit` at all, when each propeller has its own limit?" Because the demand must be limited where the controller can see it; otherwise the integral cannot know it was cut, and windup follows (§4-5).
> - **Take away** — to steer a rudderless vessel, push one side harder.

4-2. What the vessel does, measured from outside

This section answers: what can one step of yaw moment reveal about the heading?

Model `W04_C_open_loop` has no controller. The vessel runs ahead on $X_{ff} = 60$ N, and at $t = 5$ s a constant yaw moment N is added. Measured by Experiment 4-2, below:

N [N·m]	final turn rate [deg/s]	turn rate per N·m	time to 63 % of the rate [s]	heading at 40 s [deg]
5	3.05	0.610	0.36	108.2
10	5.06	0.506	0.30	178.3
20	8.14	0.407	0.24	286.0

What follows from it, one line at a time.

1. **The heading never stops.** A constant moment gives a constant turn rate, and the heading grows for as long as the moment is applied: 108°, 178°, 286° at 40 s. The heading is the running sum — the integral — of the turn rate.
2. **So the plant contains an integrator of its own.** In the language of Week 2 §2-4 it is **type 1** without any help from the controller, and by the final value theorem a proportional controller alone will therefore end **exactly** at the command. This is the opposite of Weeks 2 and 3, where P always left a gap.
3. **Holding a heading costs no moment.** Line 1 read backwards: zero moment gives zero turn rate, so a vessel already on its heading needs nothing to stay there. There is no drag and no spring on this axis to be held against — which is why line 2 comes out as it does.
4. **The turn rate answers quickly.** It is within 63 % of its final value in 0.36 s or less, so the slow part of a heading change is not the vessel's response but the **adding up** of the rate.
5. **The turn rate is not proportional to the moment.** Doubling the moment from 10 to 20 N·m raises the rate by only $61\,\%$: the yaw damping grows with the rate itself (Week 1: $N_r(1 + 10|r|)\,r$), so a quick turn meets more resistance. Unlike the surge speed of Week 3 this axis is visibly **nonlinear**, which is one more reason to tune by measurement rather than from a model.

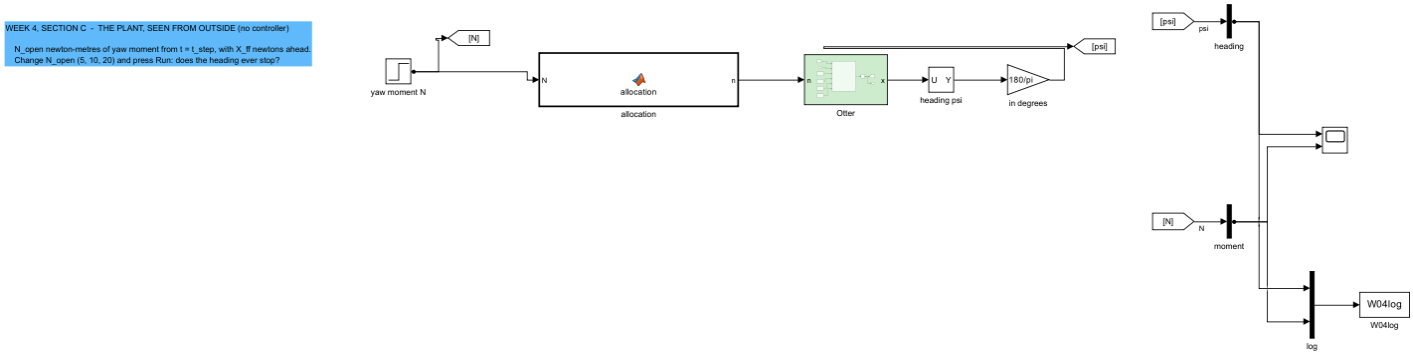
Experiment 4-2 · The heading seen from outside (10 min)

What it measures. Lines 1, 4 and 5: a heading that grows without end, a turn rate that settles within a third of a second, and a rate per N·m that falls as the moment grows.

The model. `W04_C_open_loop` — no controller. The vessel runs ahead on $X_{ff} = 60$ N and a constant yaw moment is added at $t = 5$ s.

Opening and running.

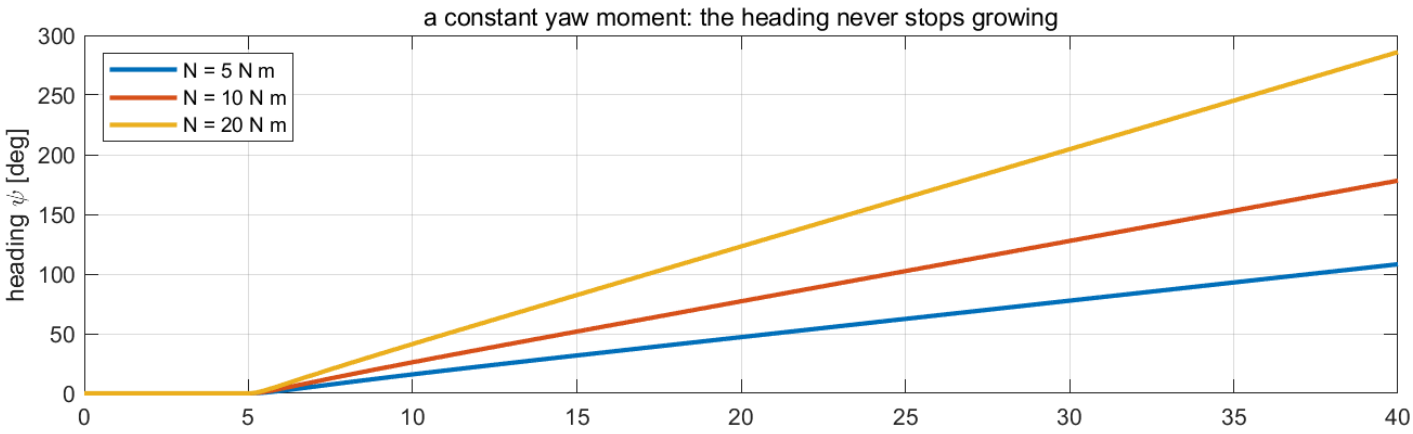
```
W04_0_setup
open_system('W04_C_open_loop')     % Run — 선수각이 멈추지 않는다 / the heading never stops
N_open = 20;                       % Run — 모멘트는 네 배, 회두율은 2.7 배뿐 / not proportional
W04_C_plant_from_outside           % 세 모멘트를 한 번에 / all three moments at once
```

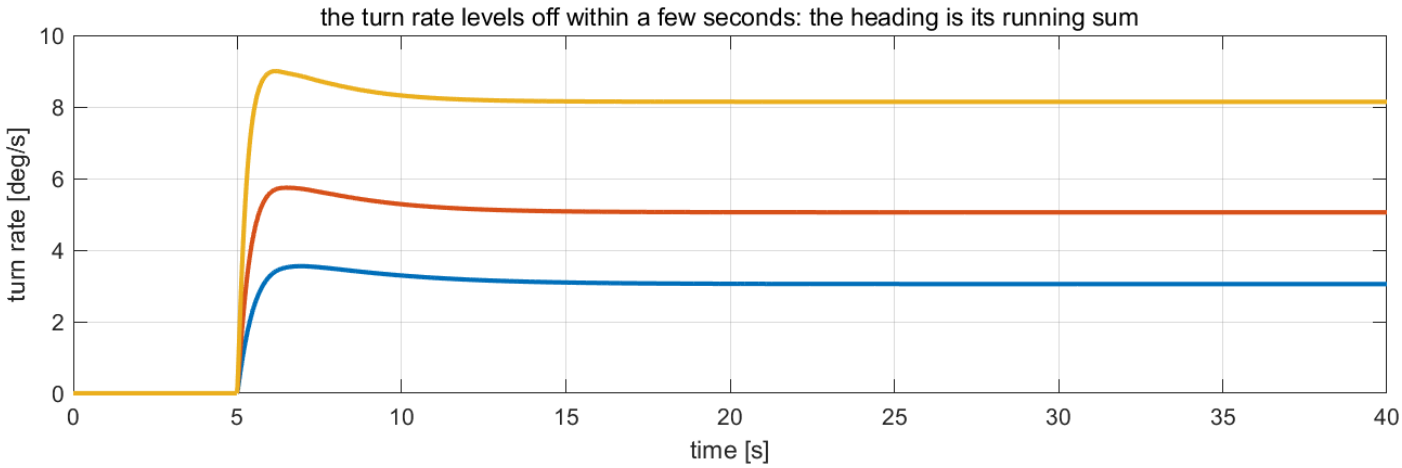


In the figure	Meaning
`yaw moment N`	a step of `N_open` N·m at $t = 5$ s
the thrust split, `Otter` , `heading psi` , `in degrees`	the plant chain of §4-1
Scope	top: the heading ψ; bottom: the moment N

Expected output:

```
  W04 Experiment 4-2  the plant seen from outside (no controller, X_ff = 60 N ahead)
    N [N m]   final turn rate [deg/s]   rate per N m   time to 63 % of the rate [s]   heading at 40 s
[deg]
    5                          3.05          0.610                          0.36
108.2
    10                         5.06          0.506                          0.30
178.3
    20                         8.14          0.407                          0.24
286.0
```





In the figure	Meaning
top	the heading after a step moment of 5, 10 and 20 N·m
bottom	the turn rate, the slope of the top curves

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, the curves are **straight and never level off**	108°, 178°, 286° at 40 s	line 1: a constant rate integrated is a heading that grows without end
bottom panel, each rate **flat** after a fraction of a second	3.05, 5.06, 8.14 deg/s	line 4: the rate is the fast part; 0.36 s to 63 %
top panel, the **slope** of each straight line	equals the flat value below it	line 1 again: the heading is the running sum of the rate
the three flat rates against 5, 10, 20 N·m	$4\times$ the moment gives only $2.7\times$ the rate	line 5: the yaw damping grows with the rate
the bottom panel **before** $t = 5$ s	zero moment, zero rate, heading unchanged	line 3: holding a heading costs nothing

What the figure says

- The heading grows without end while the turn rate levels off within a few seconds. The heading is the running sum of the turn rate.

What to try	What to watch
`N_open = 40;` Run	twice the moment of the last row gives 12.82 deg/s against 8.14 — a factor 1.57, not 2: line 5 bites harder the faster the vessel turns
`N_open = 0;` Run	the rate is 0.00 deg/s and the heading is still 0° at 40 s: line 3, holding a heading costs no moment

> ✎ **In class**
>
> - **Purpose** — contrast with Week 3's open loop: there a constant force gave a constant speed; here a constant moment gives a heading that never stops.
> - **Point to** — the straight lines in the top panel; the flat rates in the bottom one.
> - **Ask** — "What moment does it take to hold a heading?" None: a vessel on a steady heading needs no yaw moment. That is why P alone will leave no error (Experiment 4-3a).
> - **Ask** — "Why does 20 N·m give less than twice the rate of 10 N·m?" The yaw damping grows with the turn rate.
> - **Take away** — the heading is an integral; the plant brings its own.

4-3. P, I and D on the heading

This section answers: what does each term do on this axis, and how does it compare with Weeks 2 and 3?

What is observed. With the same three terms again, on this axis:

- P alone **rings** — overshoot $5.8 \to 15.9\,\%$ as K_p rises from 100 to 1000 — and yet leaves **no error at all**: the final heading is 10.000°.
- D damps it, as in Week 2 and unlike Week 3: overshoot $12.18 \to 0.39\,\%$, settling $3.26 \to 1.72$ s.
- I is not needed to reach the command, and becomes necessary only when something pushes back — a port propeller at $70\,\%$ leaves PD 0.80° off.

The heading is therefore the third of three different answers from the same controller, and the two lines of §4-2 explain all of them.

What follows from it, one line at a time.

1. **The heading can overshoot, because it is an integral.** The vessel keeps turning while the moment is removed, so the heading passes the command before the rate reaches zero — the store that Week 3's speed axis did not have.
2. **P alone leaves no error**, by §4-2 lines 2 and 3: the plant already contains an integrator, and holding a heading costs no moment. At the command the error is zero, hence $N = 0$, and zero is exactly what is needed. Compare Week 3, where holding a speed cost a steady 116 N and P therefore had to keep an error to produce it.
3. **D damps here.** The derivative of a heading error is a **turn rate**, and resisting a rate is damping, as resisting the mass's velocity was in Week 2. On the speed axis of Week 3 the same derivative was an acceleration, which is why it hurt there. The term has not changed; the axis has, for the third time.
4. **Most of the models of this week differentiate the error**, exactly as Weeks 2 and 3 did: the block `D filter` is $K_d N_f s/(s + N_f)$ fed by $\mathrm{ssa}(\psi_d - \psi)$. That is the textbook form, and it is kept in `W04_D_P` … `W04_H_tuning` so that its one weakness can be measured rather than described. **The form the rest of this course actually uses is the other one**, $-K_d r$, and the subsection "Why the derivative is better taken from the yaw rate" derives it, measures it in Experiment 4-3e, and shows what happens when its sign is wrong.
5. **I is needed only against a steady disturbance.** A propeller delivering $70\,\%$ of its thrust makes a yaw moment that never goes away; PD can oppose it only from error, and settles 0.80° off, holding 4.19 N·m. The integral finds that moment by itself — 4.15 N·m at $K_i = 20$ — which is Week 2 §2-8 line 5 once more, with a fouled propeller in place of a spring.

Term	Week 2, position	Week 3, speed	This week, heading	Measured
P	faster, rings, leaves an error	faster, never rings, leaves an error	faster, **rings**, and leaves **no error**	Experiment 4-3a: overshoot $5.8 \to 15.9\,\%$ as K_p rises $100 \to 1000$; final heading 10.000°
D	damps	makes it worse	**damps again**	Experiment 4-3b: overshoot $12.18 \to 0.39\,\%$, settling $3.26 \to 1.72$ s as K_d rises $0 \to 100$ at $K_p = 300$
I	removes the error	removes the error	removes an error only when something **pushes back**: a weak propeller	Experiment 4-3c: 0.80° left by PD with the port propeller at 70 %; removed by $K_i = 20$

Experiment 4-3a · P only (10 min)

What it measures. Lines 1 and 2: a heading that ends exactly at the command whatever the gain, and an overshoot that grows with the gain because the plant integrates.

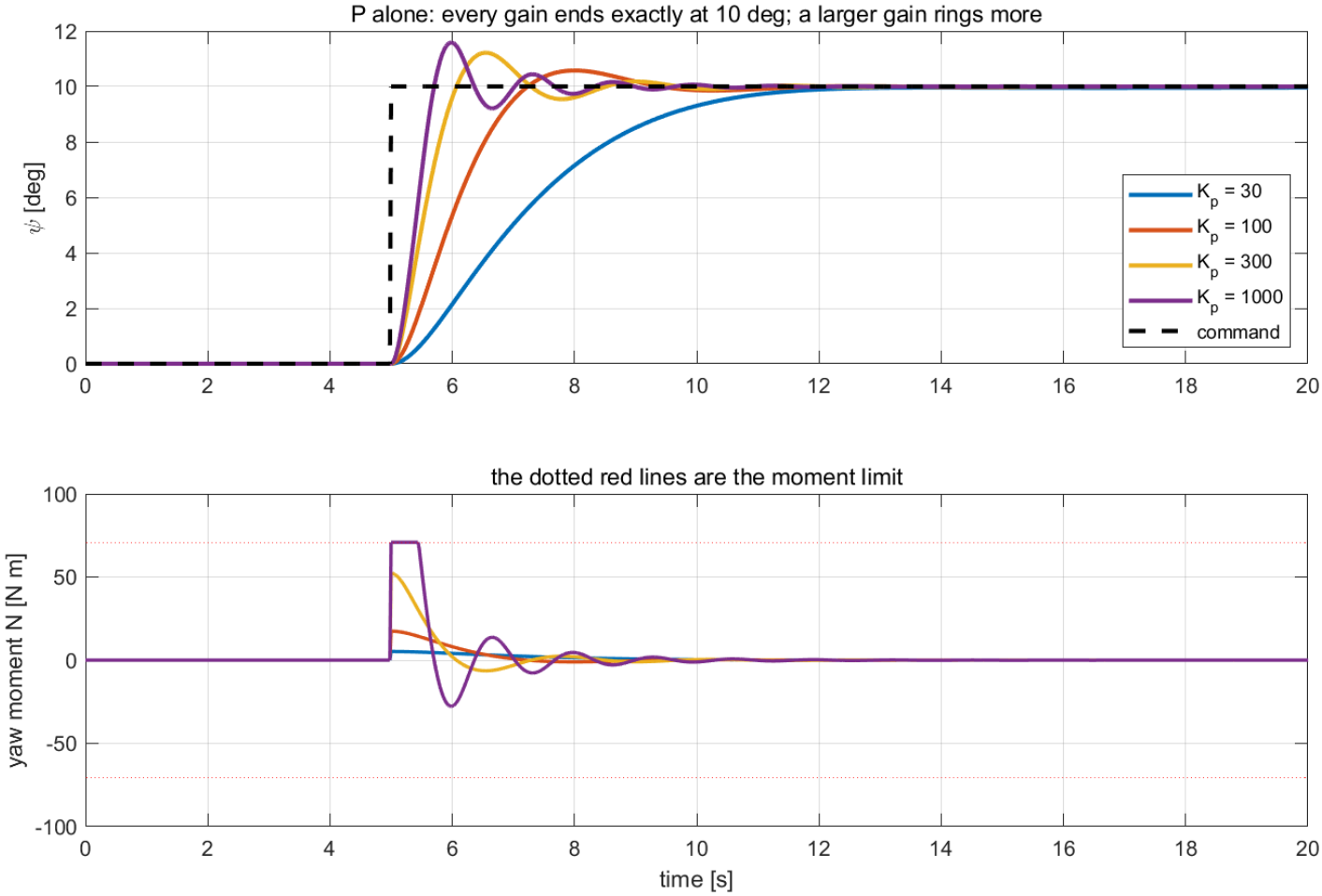
The model. `W04_D_P` — the loop of §4-1 with $K_i = K_d = 0$.

Opening and running.

```
W04_0_setup
open_system('W04_D_P')            % Run — 10.000 도에서 정확히 멈춘다 / it ends exactly at 10.000 deg
Kp = 1000;                        % Run — 빨라지고 더 넘친다 / faster, and it overshoots more
W04_D_proportional_only           % 네 게인을 한 번에 / all four gains at once
```

Expected output:

```
W04 Experiment 4-3a  P only  (psi_d = 10 deg at t = 5 s)
  Kp     final psi [deg]   overshoot %   rise [s]   settle [s]   peak |N| [N m]
  30               9.989          0.0       3.94         6.36             5.2
  100              9.997          5.8       1.50         3.96            17.5
  300              9.999         12.2       0.74         3.26            52.4
  1000            10.000         15.9       0.48         3.12            70.8
```



In the figure	Meaning
top	the heading for $K_p = 30$ to 1000 and the dashed command
bottom	the yaw moment; the dotted red lines are ± 70.85 N·m

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, where **every** curve ends	10.000°, on the dashed command	line 2: the plant's own integrator, and no moment needed to stay
top panel, the **peak above the command**	$0.0 \rightarrow 15.9\,\%$ as K_p grows	line 1: the vessel is still turning when the moment is taken away
bottom panel, the moment **at the command**	back to zero in every run	line 2 again: zero error, zero moment, and that is enough
bottom panel, the **negative** moment after the peak	present in every ringing run	line 1: the moment must be reversed to stop the turn the vessel is still doing
bottom panel at $K_p = 1000$, the **flat top**	on the 70.85 N·m line	§4-1: the moment limit, which is what §4-5 must then manage

What the figure says

- Every gain ends at 10°; a larger gain is faster and rings more. At $K_p = 1000$ the moment is on its limit and more gain buys little.

What to try	What to watch
`Kp = 30;` Run	slow but still exact: 9.989° with no overshoot at all. Line 2 does not depend on the size of K_p
`psi_step = 90; Kp = 300; T_final = 60;` Run	the moment sits on its limit for 4.24 s, and the overshoot is **smaller**, $0.91\,\%$ against $12.18\,\%$ at 10°: a saturated P loop simply turns at its largest rate, and there is no integrator to store anything. §4-5 adds one, and the same turn then overshoots by $15.6\,\%$

> ✎ **In class**
>
> - **Purpose** — the two differences from Week 3 on one plot: no error left, and ringing.
> - **Point to** — the final-heading column, all 10° to within 0.011°; the overshoot column growing with the gain.
> - **Ask** — "Week 3 at $K_p = 800$ still missed by 0.133 m/s. Why is this different?" Holding a speed needs force against drag; holding a heading needs no moment.
> - **Take away** — on an integrating axis P reaches the command; the question becomes how much it rings.

Experiment 4-3b · Add D (10 min)

What it measures. Lines 3 and 4: the derivative damping the ringing of Experiment 4-3a, and the size of what it demands at the instant of the step.

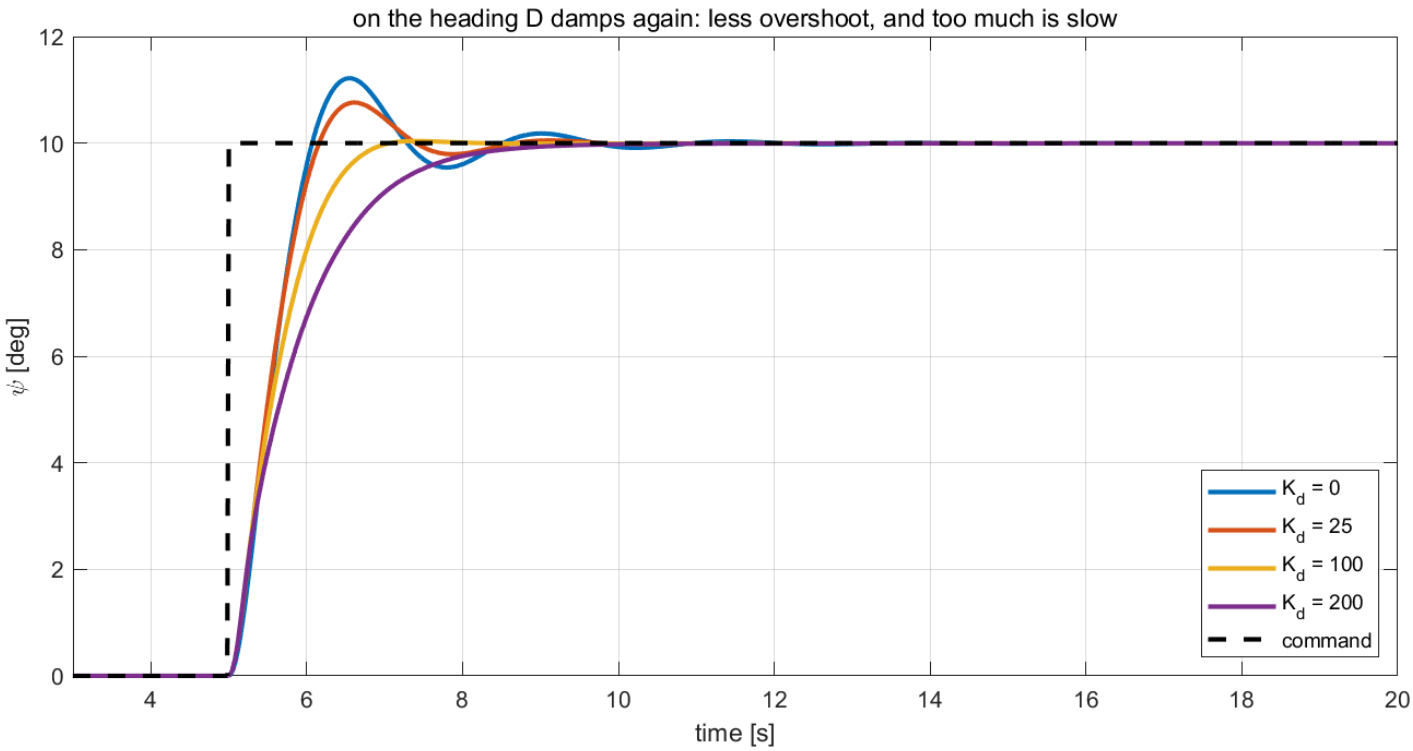
The model. `W04_E_PD` — the same loop with the `D filter` branch switched on, $K_d N_f s/(s + N_f)$ fed by the error. K_i is still zero. Where that derivative should come from is the subject of the subsection below.

Opening and running.

```
W04_0_setup
open_system('W04_E_PD')          % Run — Kd = 100: 거의 넘치지 않는다 / almost no overshoot
Kd = 0;                          % Run — 4-3a 의 울림이 돌아온다 / the ringing of 4-3a is back
Kd = 200;                        % Run — 넘침은 0, 대신 느리다 / no overshoot, but slow
W04_E_derivative                 % 일곱 값을 한 번에 / all seven gains at once
```

Expected output:

```
W04 Experiment 4-3b  P + D  (Kp = 300, psi_d = 10 deg)
  Kd     overshoot %   rise [s]   settle [s]
  0            12.18       0.74         3.26
  10           10.23       0.76         3.20
  25            7.61       0.80         2.98
  50            4.10       0.88         2.22
  100           0.39       1.12         1.72
  150           0.00       1.44         2.42
  200           0.00       1.78         3.14
```



In the figure	Meaning
traces	the heading for $K_d = 0, 25, 100, 200$ at $K_p = 300$

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
the **peak** of each trace	$12.18 \to 0.39\,\%$ as K_d grows to 100	line 3: a term against the turn rate is damping on this axis
the traces at $K_d = 150$ and 200	no overshoot, but rise 1.44 and 1.78 s	Week 2 §2-3: past $\zeta = 1$ more damping only costs speed
the **settling** column	falls to 1.72 s at $K_d = 100$, then rises	the two lines above pulling in opposite directions; the measurement picks $K_d = 100$
the moment at the instant of the step	a spike of 325.8 N·m, against a 70.85 N·m limit	line 4: the derivative of a step, $K_d N_f$ times the jump — the kick of Week 2 §2-10, and the reason for the subsection below

What the figure says

- The derivative removes the ringing; beyond $K_d = 100$ it only slows the turn.

What to try	What to watch
`Kd = 400;` Run	no overshoot at all, but the rise time is 3.24 s against 1.12 s at $K_d = 100$: the brake now outweighs the push
`Kp = 1000; Kd = 100;` Run	$5.83\,\%$ of overshoot returns, where the same K_d left $0.39\,\%$ at $K_p = 300$ — K_d is read against the K_p it damps, never on its own

> ✎ **In class**
>
> - **Purpose** — the derivative on its home ground again, after it failed on the speed loop.
> - **Point to** — the settling column: it falls to 1.72 s at $K_d = 100$ and rises again — too much damping is slow (Week 2 §2-3, $\zeta > 1$).
> - **Ask** — "The same block made Week 3 worse. What changed?" The axis: the derivative of a heading error is a turn rate, and resisting it is damping.
> - **Take away** — use D where the output can overshoot, and stop raising it where the settling time turns up.

Experiment 4-3c · A weak port propeller, and the integral (10 min)

What it measures. Line 5: a disturbance that never goes away, the error PD keeps in order to oppose it, and the integral finding the same moment without an error.

The model. `W04_F_PID` — the full loop, with `port_eff` scaling the port propeller's thrust. At `port_eff = 1` nothing is wrong and the integral has nothing to do; at `0.7` the vessel is pulled off heading for as long as it runs.

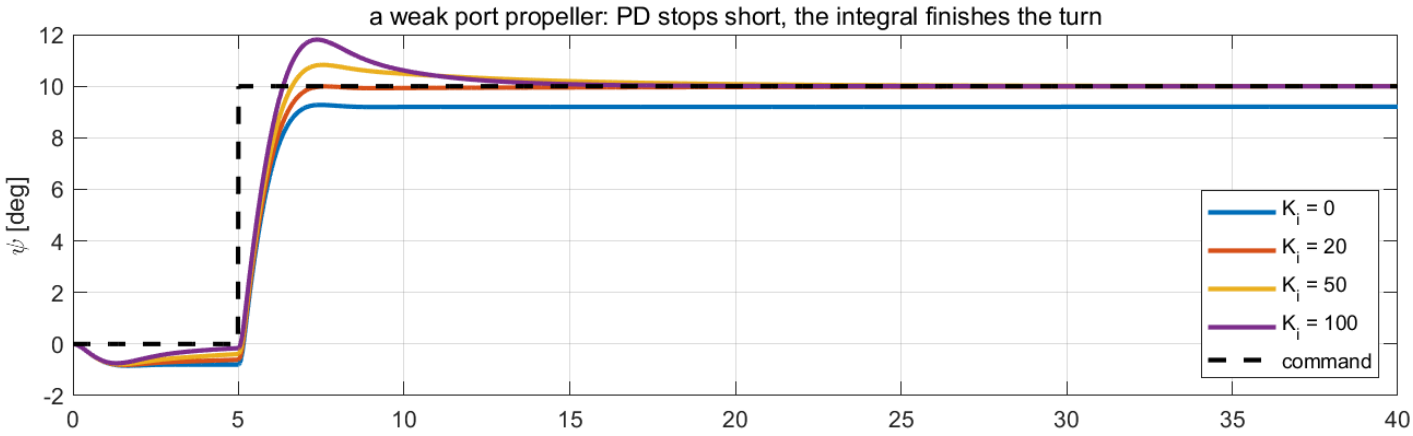
Opening and running.

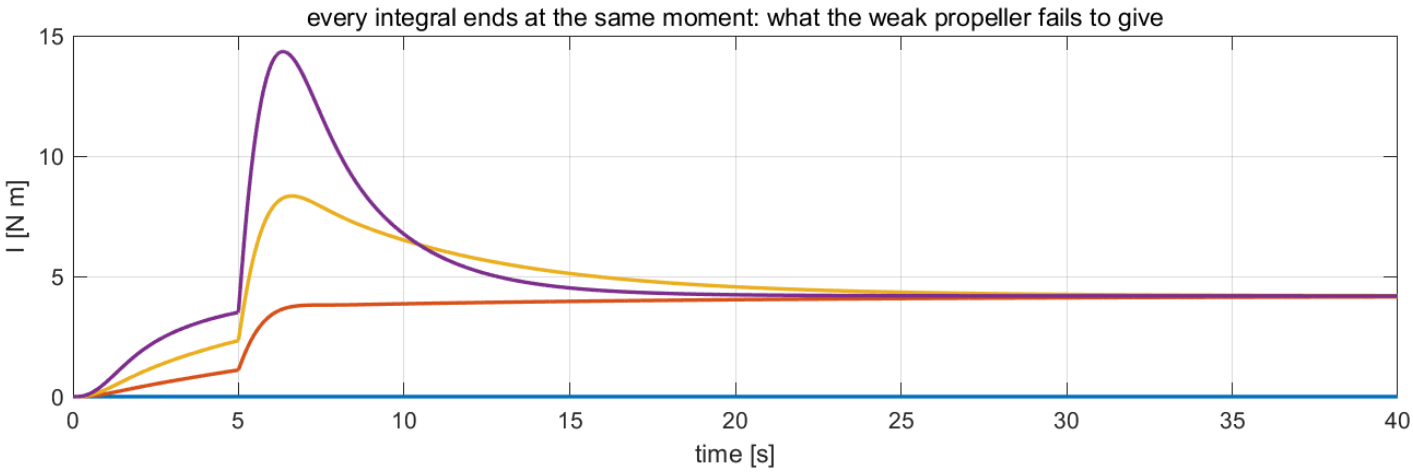
```
W04_0_setup
port_eff = 0.7; Ki = 0;          % 좌현 30 % 약함, 적분 없음 / a weak port propeller, no integral
open_system('W04_F_PID')         % Run — 9.200 도에서 멈춘다 / it stops 0.8 deg short
Ki = 20;                         % Run — 목표에 도달한다 / it reaches the command
W04_F_weak_propeller             % 두 스윕을 한 번에 / both sweeps at once
```

Expected output:

```
W04 Experiment 4-3c  1) PD only (Kp = 300, Kd = 100), port propeller at reduced efficiency
  port_eff   final psi [deg]   error left [deg]   steady N [N m]
  1                   9.999              0.001             0.00
  0.7                 9.200              0.800             4.19
  0.5                 8.490              1.510             7.90

2) add the integral, port_eff = 0.7
  Ki     final psi [deg]   overshoot %   settle [s]   I at 40 s [N m]
  0                9.200          0.00          Inf              0.00
  20               9.993          0.00         1.98              4.15
  50              10.002          8.27         9.82              4.20
  100             10.000         18.07         7.66              4.19
```





In the figure	Meaning
top	the heading with the port propeller at 70 %, for $K_i = 0$ to 100
bottom	the integral term; every curve ends at the same moment

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, the $K_i = 0$ curve	flat at 9.200°	line 5: PD can produce the needed 4.19 N·m only by keeping 0.80° of error
bottom panel, where **every** integral levels off	4.15 to 4.20 N·m, whatever K_i	Week 2 §2-8 line 5: an integrator stops where the error is zero, at the moment the fault demands
top panel, $K_i = 20$ against $K_i = 100$	$0.00\,\%$ against $18.07\,\%$ of overshoot	too much integral is late moment, on an axis that already integrates twice over
the `port_eff` table, $1 \to 0.7 \to 0.5$	steady N of $0.00 \to 4.19 \to 7.90$ N·m	line 5: the worse the propeller, the larger the moment to be held, and the larger the error PD keeps

What the figure says

- PD stops 0.80° short, because holding the heading now needs 4.19 N·m that only an error can produce. The integral supplies it and the error goes; too much integral overshoots.

What to try	What to watch
`port_eff = 1; Ki = 20;` Run	with nothing to oppose, the integral settles at 0.22 N·m and the heading ends at 10.043°: an integral is insurance, not a gain to be raised for its own sake
`port_eff = 0.5; Ki = 20;` Run	the same K_i still reaches the command, 9.947°, now holding 7.64 N·m — the term finds whatever the fault costs

> ✎ **In class**
>
> - **Purpose** — show when the integral is needed on an axis that already integrates: when something pushes back.
> - **Point to** — the steady N column: 4.19 N·m with PD, the same 4.15–4.20 N·m in the integral once $K_i > 0$.
> - **Ask** — "Why does the weak propeller make a yaw moment at all?" Both propellers are asked for 30 N; the port one gives 21 N, so the starboard side pushes harder and the vessel turns to port.
> - **Take away** — the integral finds whatever steady effort the plant demands, whether drag (Week 3) or a fault (here).

Why the derivative is better taken from the yaw rate

This subsection answers: the autopilot must brake the turn — why measure the **yaw rate** rather than differentiate the heading error?

The two forms. Both are called "the D term", and they are not the same expression:

$$\text{error form:}\quad N = K_p\,e + K_d\,\dot{e}, \qquad \text{rate form:}\quad N = K_p\,e - K_d\,r, \qquad e = \operatorname{ssa}(\psi_d - \psi)$$

Why they are usually equal. Differentiating the error gives $\dot{e} = \dot{\psi}_d - \dot{\psi} = \dot{\psi}_d - r$. **Whenever the command is not moving**, $\dot{\psi}_d = 0$ and the two forms are identical, term for term. Everything the derivative does while the vessel settles on a fixed heading is the same in both.

Where they part, one line at a time.

1. They differ **only** through $\dot{\psi}_d$: the error form adds $K_d\,\dot{\psi}_d$, and nothing else.
2. A step command makes $\dot{\psi}_d$ an impulse. Filtered as §2-10 filters it, the term does not go to infinity but starts at $K_d N_f$ times the jump: $100 \times 20 \times 0.1745\ \text{rad} = 349$ N·m for the 10° step, against a moment limit of 70.85 N·m. **The rate form cannot do this**, because r is a velocity of a vessel with inertia and cannot jump.
3. That spike is spent on the actuator, not on the vessel: it is cut off by the limit within a few hundredths of a second, and §4-5 records the windup it causes even in a 10° turn.
4. **The seam makes it worse.** When a command crosses $\pm 180^\circ$, the wrapped error jumps by a full 2π — that is what `ssa` does, and §4-6 shows why it must. Differentiating that jump asks for $K_d N_f\,2\pi = 12\,566$ N·m, some 177 times the limit, for a turn of a few degrees. The rate form is untouched by it.
5. **The sensor decides it too.** A vessel carries a **gyroscope**, and r is what it reports directly. The error form has no measurement of $\dot{\psi}$: it must compute one by differentiating the heading, and a derivative multiplies every frequency by ω (§2-10) — so it amplifies heading noise, and then needs the filter N_f, whose phase lag costs some of the damping it was added for.
6. The price of the rate form is that it does not anticipate a **moving** command; a ramped command is followed with a small lag. A reference model that supplies ψ_d together with $\dot{\psi}_d$ removes even that, and is the subject of Week 8.

Why the sign must be negative. The rate term is written $-K_d r$ with $K_d > 0$, and the minus is not a convention — it is the whole of the term's purpose. Put the law into the yaw equation of §4-2 line 5:

$$M_{66}\,\dot{r} \;=\; N \;-\; \underbrace{|N_r|\big(1 + 10|r|\big)\,r}_{\text{the hull's own damping}}, \qquad N = K_p\,e - K_d\,r$$

and with $e = \psi_d - \psi$ and $r = \dot{\psi}$ this is

$$M_{66}\,\ddot{\psi} + \Big[\underbrace{|N_r|\big(1 + 10|r|\big)}_{\text{the hull's}} + \underbrace{K_d}_{\text{the controller's}}\Big]\dot{\psi} + K_p\,\psi = K_p\,\psi_d$$

Symbol	Quantity	Value · source
M_{66}	yaw inertia, added mass included	42.65 kg·m² · `otter.m`, Week 1 §1-9
N_r	linear yaw damping	-42.65 N·m per rad/s · $-M_{66}/T_{\text{yaw}}$, $T_{\text{yaw}} = 1$ s
K_d	the rate gain	100 N·m per rad/s

7. **K_d sits in the same bracket as the hull's own damping, and adds to it.** That is what "damping" means here, and it is why the term is subtracted from the moment: a moment **against** the turn is a moment that removes energy from it.
8. Flip the sign and the bracket becomes $|N_r|(1 + 10|r|) - K_d$. The controller now **subtracts** from the hull's damping, and with $K_d = 100$ against $|N_r| = 42.65$ the total is **negative** at low turn rates: the loop puts energy in on every swing. A brake wired backwards is an accelerator.
9. It does not fly apart, and the reason is Week 1's nonlinear factor. The bracket returns to zero when $|N_r|(1 + 10|r|) = K_d$, that is at

$$|r| = \frac{1}{10}\left(\frac{K_d}{|N_r|} - 1\right) = \frac{1}{10}\left(\frac{100}{42.65} - 1\right) = 0.1345\ \text{rad/s} = 7.70\ \text{deg/s}$$

so the vessel grows its oscillation until the turn rate reaches about that value and then holds it — **a limit cycle, not a divergence**. Experiment 4-3e measures 7.23 deg/s.

What MSS does. Not one vessel in the toolbox differentiates the error. Three of them steer by the rate form, and they are the reference this course follows:

MSS file	The line	What it is
`VESSELS/SIMclarke83.m`	`M(3,3)*( Kp*ssa(psi_ref-eta(3)) - Kd*nu(3) )`	a PD heading autopilot; $\nu(3)$ is r
`VESSELS/SIMremus100.m`	`delta_r = -Kp_psi*ssa( x(12)-psi_d ) - Kd_psi*x(6)`	the same, with the error written the other way round; $x(6)$ is r
`VESSELS/SIMrig.m`	`tau3 = -R'*(Kp*eta3 + Ki*z3) - Kd*nu3`	3-DOF DP: P and I on the **position error**, D on the measured **velocity ν**

`SIMrig` is the one worth reading twice. The proportional and integral terms act on an error, and the damping term acts on a measured velocity — they are not three derivatives of one signal. That is exactly the structure Week 8 §8-3 builds and Week 9 assembles.

> **Two ways to write the same gain**
>
> Some texts — including the companion Capstone course — fold the minus into the gain and write $N = K_p e + K_{d,\psi}\,r$ with $K_{d,\psi} < 0$. This course keeps $K_d > 0$ and writes the minus in the law, so that every gain in every week is positive. They are the same controller; $K_{d,\psi} = -K_d$.

	error form, $K_d\dot{e}$	rate form, $-K_d r$
on a fixed command	identical	identical
at a step in ψ_d	$\mathbf{349}$ N·m demanded	$\mathbf{-1.1}$ N·m: nothing happens
at a $\pm 180^\circ$ seam crossing	$\mathbf{12\,566}$ N·m	nothing happens
sensor needed	a differentiated heading, and a filter	the gyro, as it is
when the command ramps	anticipates it	lags slightly behind
used in	this week's models, so that the kick can be measured	**Week 5**, where guidance moves ψ_d at every waypoint

Experiment 4-3d · The kick, and what the rate form would have asked for (5 min)

What it measures. Lines 2 and 5: the D term actually demanded at the 10° step, and what $-K_d r$ would have demanded in the same run.

The run. The log of Experiment 4-3b, on `W04_E_PD` — no new model and no new simulation. Both forms are read from the same run, because both see the same vessel.

Opening and running.

```
W04_0_setup
R = W04_read('W04_E_PD', 'Kp', 300, 'Kd', 100);
r = gradient(deg2rad(R.psi), R.t);        % 기록된 선수각에서 회두율 / yaw rate from the log
[max(abs(R.D))  max(abs(Kd*r))  N_max]    % 오차 미분 / 회두율 되먹임 / 한계
```

Expected output:

```
ans =

  325.7948   20.9746   70.8488
```

Reading the output.

Printed	What it is	Which line it shows
$\mathbf{325.79}$	the largest D term the models actually demand	line 2: the kick, $\mathbf{4.6}$ times the moment the propellers can produce
$\mathbf{20.97}$	the largest $\lvert K_d r \rvert$ in the same run	line 2: the rate form never asks for more than the vessel is doing — the peak yaw rate is $\mathbf{12.02}$ deg/s, and $\mathbf{100 \times 0.2097 = 21.0}$ N·m
$\mathbf{70.85}$	the moment limit	the line both are judged against

- At the instant of the step the two are $\mathbf{325.8}$ N·m and $\mathbf{-1.1}$ N·m. A second later, with the command no longer moving, they agree — which is line 1.

What to try	What to watch
`Nf = 5;` then rerun the two lines	the kick falls to 85.8 N·m, close to the $K_d N_f \times 0.1745 = 87.3$ the formula gives, and **still above the limit**: filtering reduces the kick but cannot remove it, because it is the command that jumped. The rate form meanwhile rises slightly, to 26.4 N·m, because the slower filter lets the turn run harder
`psi_step = 90;` `T_final = 60;` then rerun	the kick grows with the step: 2932 N·m, against $K_d N_f \times 1.571 = 3142$ from the formula. The rate form reaches 33.3 N·m — still within what the hull can be asked for

Experiment 4-3e · The three D terms, each one actually steering (10 min)

What it measures. Lines 1, 7 and 8: that the two forms track alike while the command is fixed, what each one demands, and what the wrong sign does when it is genuinely in the loop.

Experiment 4-3d read both forms out of **one** run. This one closes the loop on each in turn, so the vessel really is steered by the term under test — the difference matters for the third form, which changes the trajectory completely and therefore cannot be read off a run it did not fly.

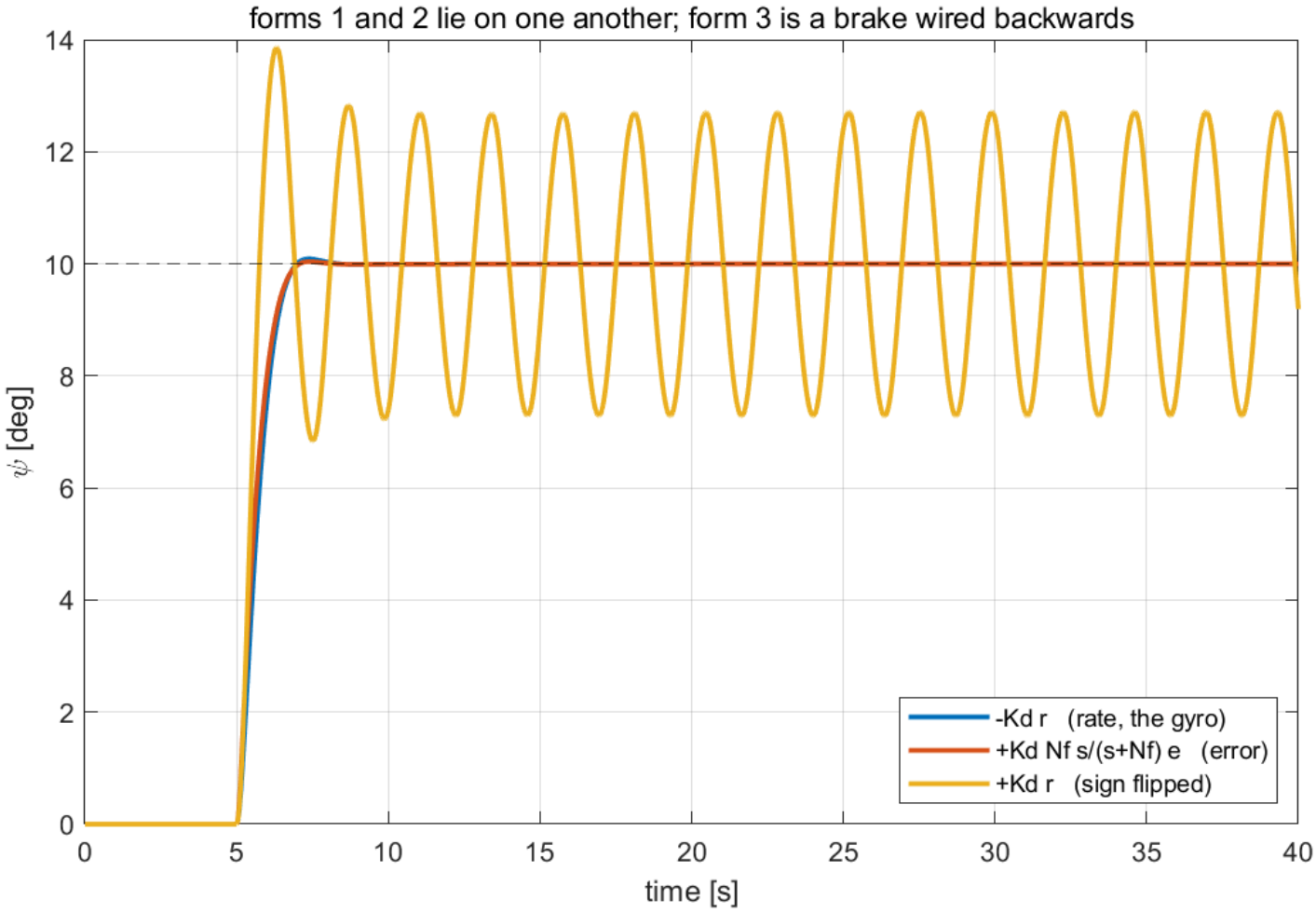
The model. `W04_I_rate` — the loop of Experiment 4-3b with the single `D filter` replaced by three branches into a `Multiport Switch`, selected by `d_form`: $-K_d r$ from the gyro, the filtered $K_d \dot{e}$, and $+K_d r$. Everything else — vessel, gains, command, solver — is shared, so the three runs differ in that one branch and in nothing else.

Opening and running.

```
W04_0_setup
open_system('W04_I_rate')     % Run — d_form = 1, 회두율 되먹임 / the rate form
d_form = 2;                   % Run — 오차를 미분. 궤적은 거의 같다 / the error form; nearly the same tr
ack
d_form = 3;                   % Run — 부호를 뒤집는다. 멈추지 않는다 / the sign flipped; it never stops
W04_I_rate_feedback           % 세 가지를 한 번에 / all three at once
```

Expected output:

```
W04 Experiment 4-3e  where the D term comes from  (Kp = 300, Kd = 100, psi_d = 10 deg)
  d_form   D term                          overshoot   rise [s]   settle [s]   max |D| [N m]
    1      -Kd r   (rate, the gyro)           0.90 %      1.16    1.74          17.03
    2      +Kd Nf s/(s+Nf) e   (error)        0.39 %      1.12    1.72         325.79
    3      +Kd r   (sign flipped)            38.59 %      0.50   never          29.46
  the wrong sign settles into a limit cycle of 7.23 deg/s;  |N_r|(1+10|r|) = Kd predicts 7.70
```



Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
blue against red	one line, to the width of the stroke	the two forms are the same term while $\dot{\psi}_d = 0$
the first two rows of the printed table	0.90 against $0.39\,\%$, 1.74 against 1.72 s	the same again, measured: what separates them is not the tracking
the `max |D|` column, first two rows	17.03 against 325.79 N·m, a factor of 19, against a limit of 70.85	line 2: the error form differentiates a step, and r belongs to a vessel with inertia and cannot jump
the yellow trace	it reaches the command and then never stops crossing it	line 8: the bracket is negative, so every swing gains energy
the line printed below the table	the swing settles at 7.23 deg/s, against 7.70 predicted	line 9: the hull's own nonlinear damping catches it there
the `settle` column, third row	`never` — it is still swinging at 40 s	line 8: nothing in the loop is removing energy at low rates

What the figure says

- Two of these three choices steer the vessel the same way and ask the propellers for numbers that differ by a factor of nineteen. The third is the same gain with one sign wrong, and it turns the brake into an accelerator.

What to try	What to watch
`psi_step = 90; T_final = 80;` then rerun	the two good forms are indistinguishable — overshoot $0.00\,\%$, settling 5.58 and 5.60 s — while the demand parts by a factor of 88: 33.27 against 2932.15 N·m
`d_form = 3; Kd = 30;` Run	$30 < \lvert N_r \rvert = 42.65$, so line 9's formula returns a **negative** rate — there is no crossing, the bracket is positive everywhere, and the vessel settles in 6.42 s with $18.45\,\%$ of overshoot. The wrong sign is survivable only while it is smaller than the damping it is fighting
`d_form = 3; Kd = 200;` Run	the limit cycle grows with the gain: $79.88\,\%$ of overshoot and a turn rate of 18.84 deg/s, against the $\frac{1}{10}(200/42.65 - 1) = 0.369$ rad/s $= 21.14$ deg/s line 9 predicts

> ✎ **In class**
>
> - **Purpose** — settle where the D term comes from, and why its sign is the whole of its meaning.
> - **Point to** — the blue and red traces lying on one another, then the log axis beside them.
> - **Ask** — "If the two track identically, why does the choice matter?" It is not the vessel that notices; it is the actuator, and the seam of §4-6.
> - **Take away** — differentiate nothing that a sensor already measures, and check the sign of a damping term against the damping the hull already has.

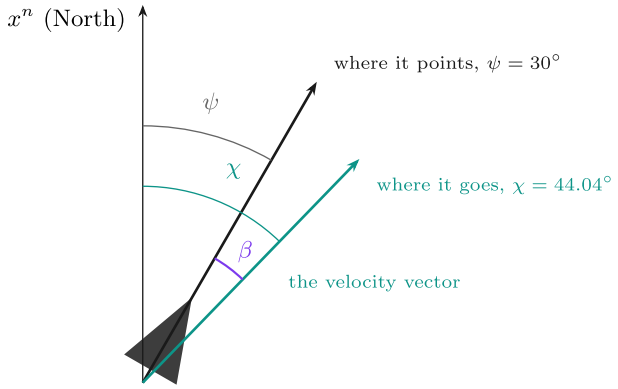
4-4. Heading is not course — the crab angle

This section answers: does a vessel go where it points?

- A heading controller regulates where the vessel **points**. It does not regulate where the vessel **goes**, and those are two different directions.

Heading, course and the crab angle

Where the vessel **points** and where it **goes** are two different directions. A heading controller regulates the first one only.



three angles, one identity

$$\chi = \psi + \beta$$

ψ heading — the hull's x axis, from North
χ course over ground — the velocity, from North
β crab angle — the gap between them, $\beta = \operatorname{atan2}(v, u)$

With $u = 2.0$ and $v = 0.5$: $\beta = 14.04°$ and $\chi = 44.04°$.

β is non-zero whenever the sway velocity v is — in a turn, in a current, in a beam wind. **It is not a disturbance to be rejected; it is what the hull is doing.** Steering ψ while travelling along χ is the permanent path error that Week 5 has to remove, and §5-5 measures it as $\Delta \tan \beta$.

In the figure	Meaning
orange ray	the heading ψ — the direction x_b points
blue ray	the course χ — the direction the velocity actually goes
green arc	the crab angle β, the gap between them

$$\beta = \operatorname{atan2}(v,\ u), \qquad \chi = \psi + \beta$$

Symbol	Definition	Unit
ψ	heading, from North to x_b	rad
χ	course over ground, from North to the velocity vector	rad
β	crab angle	rad
$u,\ v$	surge and sway velocity (Week 1)	m/s

- With the example of Week 1, $u = 2.0$ m/s and $v = 0.5$ m/s at $\psi = 30^\circ$: $\beta = \operatorname{atan2}(0.5,\ 2.0) = 14.04^\circ$ and $\chi = 44.04^\circ$.
- The crab angle is non-zero whenever $v \neq 0$ — in a turn, in a current, in a beam wind. It is what the hull is doing, not an error to be removed. Week 1 measured $\beta = -20.3^\circ$ in a turning run of a vessel with no sway actuation at all.
- Fossen writes $\beta = \arcsin(v/U)$; for $u > 0$ it is the same number. The form $\operatorname{atan2}(v, u)$ remains correct going astern.
- This week regulates ψ and lets χ fall where it may. **Week 5 cannot**: a path-following law that steers the heading while the vessel travels along the course leaves a cross-track error of the order of β times the look-ahead distance.

Experiment 4-4 · The crab angle, from two velocities (3 min)

What it measures. The formula above, on the numbers Week 1 measured. There is no model here: the crab angle is a property of the velocity, and two lines in the Command Window are the whole calculation.

Opening and running.

```
u = 2.0;  v = 0.5;  psi = 30;          % 1주차의 값 / the Week 1 example
beta = atan2d(v, u)                    % 크랩각 / the crab angle
chi  = psi + beta                      % 침로 / the course over ground
```

Expected output:

```
beta =

   14.0362


chi =

   44.0362
```

What to try	What to watch
`v = -0.5;`	$\beta = -14.04^\circ$: sway to port carries the vessel to port of where it points
`v = 0;`	$\beta = 0$ and $\chi = \psi$ — the only case in which a heading controller is also a course controller
`u = -2.0; v = 0.5;`	$\beta = 165.96^\circ$: going astern, $\operatorname{atan2}$ still returns the true direction of travel, where $\arcsin(v/U)$ would not

- The vessel of this week is never asked for a particular χ, so the crab angle is only observed. Week 5 steers a **path**, and then β becomes the standing error that ILOS has to remove.

4-5. The tuning order, applied to the heading

This section answers: how are the gains chosen for this axis, without a model — and what does the moment limit add?

The procedure of Week 2 §2-12. The requirement: a 10° turn inside 2 % within 3 s with overshoot below 5 %; no heading error with a weak propeller; no windup in a 90° turn.

Step	What is done	What was measured	Decision
0	requirement; is the plant well behaved?	§4-2: stable turn rate, integrating heading, mildly nonlinear	proceed
1	P only; scale of K_p from the units	a 10° error is 0.175 rad; $K_p = 300$ asks 52 N·m of the 70.85 available	$K_p = 300$
2	read the P response	overshoot $12.2\,\%$, settling 3.26 s: it rings	add D
3	raise K_d while the response improves	$K_d = 25$: $7.61\,\%$, 2.98 s; 50: $4.10\,\%$, 2.22 s; 100: $0.39\,\%$, 1.72 s; 150: $0\,\%$, 2.42 s	$K_d = 100$, the fastest settling
4	add I if something pushes back	a 70 % port propeller leaves 0.80°; $K_i = 20$ removes it, settling 1.98 s	$K_i = 20$
5	look at the moment: a big turn	a 90° turn saturates the moment; the back-calculation gain K_b decides what the integral does meanwhile	below

The moment limit and K_b. Measured by Experiment 4-5, below, settling time and overshoot for three turns:

K_b	10° turn	weak propeller, 10°	90° turn
0 (no anti-windup)	12.42 s, 4.35 %	1.98 s, 0 %	37.12 s, **15.57 %**
0.05	7.28 s, 3.20 %	2.18 s, 0 %	24.14 s, 6.38 %
0.1	**2.64 s, 2.06 %**	9.38 s, 0 %	**5.80 s, 0 %**
1	33.68 s, 0 %	35.00 s or more, 0 %	48.50 s, 0 %

- **Too little K_b: windup.** While the moment is on its limit the integral keeps charging — to 79.7 N·m in the 90° turn — and the vessel overshoots by $15.6\,\%$ and takes 37 s to come back. Even the 10° turn winds up a little: the derivative kick at the step touches the limit for a moment.
- **Too much K_b: the integral is dragged down.** While saturated, back-calculation pulls the integral towards the value that makes the demand equal to the limit. With $K_p e$ alone asking for several hundred N·m, that value is far below zero — -249.5 N·m at $K_b = 1$ — and the vessel stalls short of its heading for tens of seconds. This is the $I^\star$ of Week 2 §2-11, on the vessel.
- $K_b = 0.1$ is the compromise: the 90° turn behaves as if there were no integral at all (5.80 s against 5.60 s for PD alone), the 10° turn meets the requirement, and the weak propeller is still corrected, more slowly.

The result, $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$, is the default in `W04_0_setup.m`. No transfer function of the vessel was written down.

Experiment 4-5 · The tuning order, and a big turn (15 min)

What it measures. The five steps of the table above, and then the one number that only a saturating turn can decide: K_b, measured on three turns at once.

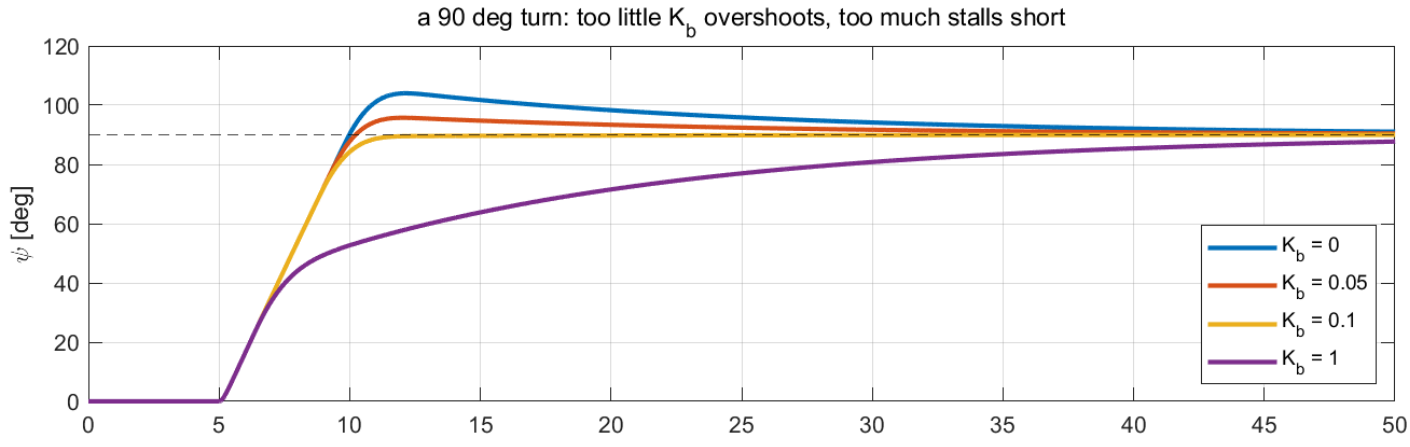
The model. `W04_H_tuning` — the full loop with the moment limit and the back-calculation path. The 90° turn is what drives the moment onto its limit for long enough to matter.

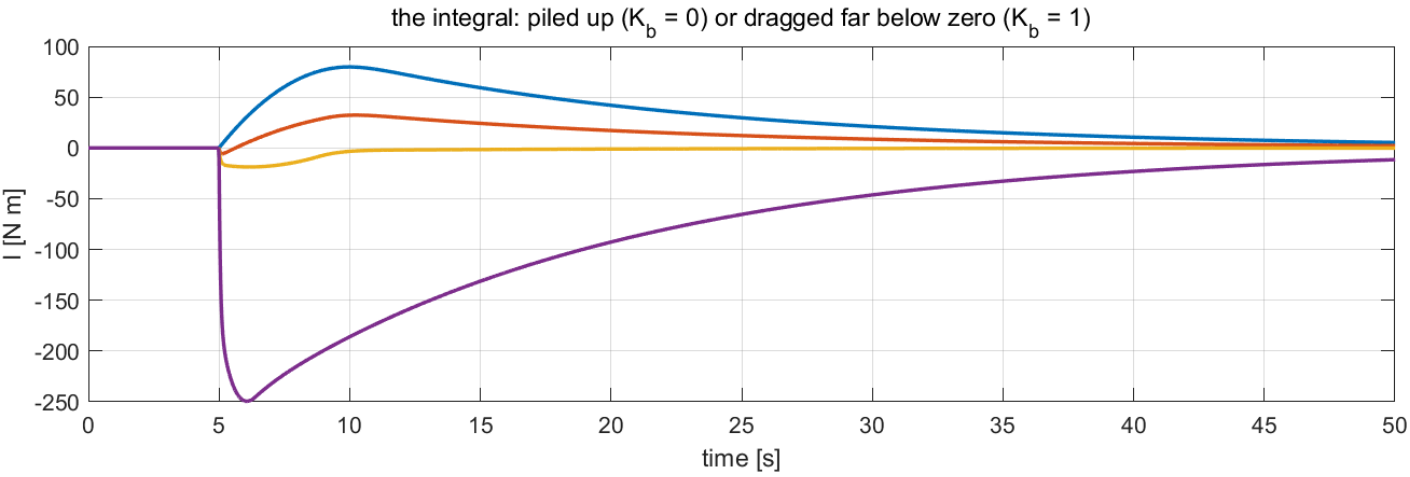
Opening and running.

```
W04_0_setup
psi_step = 90; T_final = 60;
open_system('W04_H_tuning')        % Run — Kb = 0.1: PD 만큼 깔끔하다 / as clean as PD alone
Kb = 0;                            % Run — 적분이 쌓여 15.6 % 넘친다 / windup, 15.6 % overshoot
Kb = 1;                            % Run — 너무 끌어내려 한참 못 미친다 / dragged down, and slow
W04_H_tuning_by_hand               % 다섯 단계와 Kb 세 선회를 한 번에 / all five steps and three turns
```

Expected output:

```
  W04 Experiment 4-5  the tuning order on the heading
    1-2  Kp = 300 (10 deg asks 52 N m of the 70.8 available): overshoot 12.2 %, settle 3.26 s: it ring
s, so D
    3    Kd = 25: 7.61 %, 2.98 s   50: 4.10 %, 2.22 s   100: 0.39 %, 1.72 s   150: 0.00 %, 2.42 s   ->
Kd = 100 (fastest settling)
    4    weak port propeller (0.7): PD leaves 0.80 deg; Ki = 20 removes it: 0.007 deg left, settle 1.9
8 s
    5    the moment limit: Kb against three turns (settle [s], overshoot %)
         Kb      10 deg turn        weak propeller     90 deg turn
         0       12.42 s   4.35 %    1.98 s   0.00 %   37.12 s  15.57 %
         0.05     7.28 s   3.20 %    2.18 s   0.00 %   24.14 s   6.38 %
         0.1      2.64 s   2.06 %    9.38 s   0.00 %    5.80 s   0.00 %
         1       33.68 s   0.00 %   35.00 s   0.00 %   48.50 s   0.00 %
         PD alone, 90 deg turn: 5.60 s, 0.00 %
         final: Kp = 300, Kd = 100, Ki = 20, Kb = 0.1
```





In the figure	Meaning
top	the heading in a 90° turn for $K_b = 0, 0.05, 0.1, 1$
bottom	the integral term

Reading the figure against the steps.

Where to look	What is there	Which step or rule it shows
top panel, all four curves **before** about 8 s	one line	the moment is on its limit and the vessel turns as fast as it can, whatever the integral is doing
bottom panel, the $K_b = 0$ curve **climbing**	to 79.7 N·m	step 5, windup: the error is large for seconds, and nothing drains the store
top panel, the $K_b = 0$ curve **past** 90°	$15.57\,\%$, and 37 s to return	Week 2 §2-11: the store can be spent only by going past the command
bottom panel, the $K_b = 1$ curve **diving**	to -249.5 N·m	Week 2 §2-11's I^*: with $K_p e$ asking hundreds of N·m, the value that cancels it is far below zero
top panel, the $K_b = 1$ curve **stalling** short	tens of seconds	the same: that store has to be undone before the heading can finish
top panel, the $K_b = 0.1$ curve	5.80 s, no overshoot — PD alone gives 5.60 s	step 5's compromise: the integral is neither stored nor over-drained

What the figure says

- With too little K_b the integral piles up and the vessel overshoots; with too much it is dragged far below zero and the vessel stalls short. $K_b = 0.1$ turns as cleanly as PD alone.

What to try	What to watch
`psi_step = 10; Kb = 0;` Run	even a small turn winds up a little: the moment touches its limit for an instant and the settling grows to 12.42 s
`Ki = 0; psi_step = 90;` Run	with no integral there is nothing to wind up, and K_b makes no difference at all — the remedy exists only because the term does

> ✎ **In class**
>
> - **Purpose** — the full procedure, and one decision that cannot be made without looking at the actuator.
> - **Point to** — the blue integral rising to 80 N·m and the purple one diving to -250 N·m, while the heading curves separate from about 8 s on.
> - **Ask** — "Why does a larger K_b make things worse here, when it helped in Week 3?" In a big turn $K_p e$ alone asks for several hundred N·m; back-calculation drags the integral towards the value that would cancel it, far below zero, and that has to be undone afterwards.
> - **Ask** — "Why is the weak-propeller case slower at $K_b = 0.1$?" The derivative kick at the step touches the limit, back-calculation pulls the integral down, and the integral must then climb to the 4.19 N·m the fault needs.
> - **Take away** — the back-calculation gain is a trade-off, and it is chosen by running the turns that matter.

4-6. The wrap

This section answers: why is the heading error passed through `ssa`?

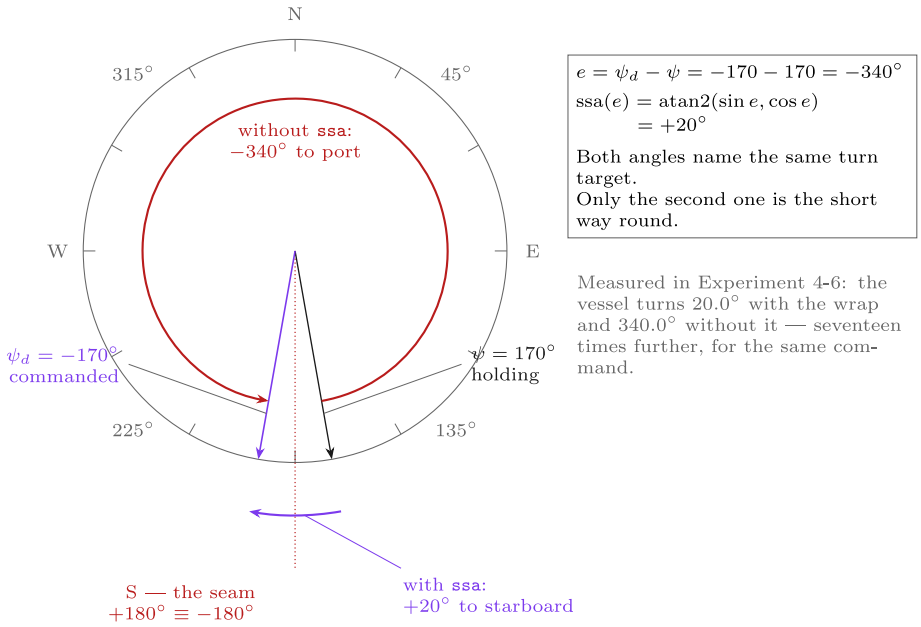
What is observed. The vessel holds 170° and is commanded to -170°. With one block removed from the error path it turns **all the way round the other way**: 340° instead of 20°.

What follows from it, one line at a time.

1. An angle and that angle plus 360° are the same heading, so the difference of two angles is only defined up to 360°. The raw subtraction $-170 - 170 = -340^\circ$ is arithmetically correct and physically absurd.
2. A controller acts on the number it is given: -340° asks for a 340° turn to port, and the vessel obeys.
3. The remedy is to map the error onto the shortest equivalent angle, $\mathrm{ssa}(e) = \mathrm{atan2}(\sin e,\ \cos e)$, which returns a value in $(-180^\circ,\ 180^\circ]$ — here $+20^\circ$. The sine and the cosine are unchanged by adding 360°, which is exactly why they are used.
4. This is not an edge case. Week 5's guidance produces commanded headings anywhere in $(-180^\circ,\ 180^\circ]$, and a mission of a few legs crosses the seam routinely.

Why the heading error is wrapped

The vessel holds $\psi = 170^\circ$ and is commanded to $\psi_d = -170^\circ$. The two headings are 20° apart, on either side of the $\pm 180^\circ$ seam.



Measured in Experiment 4-6: the vessel turns 20.0° with the wrap and 340.0° without it — seventeen times further, for the same command.

In the figure	Meaning
the grey circle	every heading, measured from North clockwise, as the vessel's compass reads it
the dotted red line at S	the seam, where $+180^\circ$ and -180° are the same direction and the number jumps
the black ray	where the vessel is pointing, $\psi = 170^\circ$
the violet ray	where it is asked to point, $\psi_d = -170^\circ$, which is 190°
the violet arc outside the circle	the turn the wrapped error asks for: $+20^\circ$ to starboard
the red arc inside the circle	the turn the raw subtraction asks for: -340° to port, the whole circle but for 20°
the box on the right	the two numbers, and the one line that turns the second into the first

Reading the figure, one angle at a time.

Step	The number	What the controller does with it
the vessel holds	$\psi = 170^\circ$	nothing yet
the command arrives	$\psi_d = -170^\circ$	the same direction as 190°; a compass cannot tell them apart
the raw subtraction	$e = -170 - 170 = -340^\circ$	a large **negative** error: turn hard to **port**, and keep turning for 340°
wrapped	$\mathrm{ssa}(-340^\circ) = +20^\circ$	a small **positive** error: ease 20° to **starboard**
after the turn	190°, or -170°	the same heading either way — only the route differed

- **What `ssa` does, in one sentence.** It leaves every error under 180° exactly as it is, and replaces every error over 180° by the equivalent turn in the other direction. It never changes the heading that is being asked for — only the route taken to it.
- **Why $\mathrm{atan2}(\sin e, \cos e)$ and not an `if`.** Adding or subtracting 360° changes neither $\sin e$ nor $\cos e$, so the pair $(\sin e, \cos e)$ already **is** the wrapped angle; `atan2` merely reads it back. One expression covers every case, including $e = \pm 540^\circ$ and beyond, with no branch to get wrong.
- **What it costs.** Nothing in steady running: for $|e| < 180^\circ$ the output equals the input, so every gain tuned in §4-3 is unaffected. It only ever acts when the seam is crossed.
- **The one thing it must come before.** The wrap belongs in front of **every** gain, and the models put it there: `e` → `ssa` → `Kp`, `D filter`, `Ki`. A 340° error reaching the integrator would store a turn the vessel is not going to make.

Measured by Experiment 4-6, below, the vessel at 170° commanded to -170° at $t = 20$ s:

`use_ssa`	turn executed after 20 s
1	20.0°
0	340.0°, the other way

- Seventeen times further, for the same commanded heading.

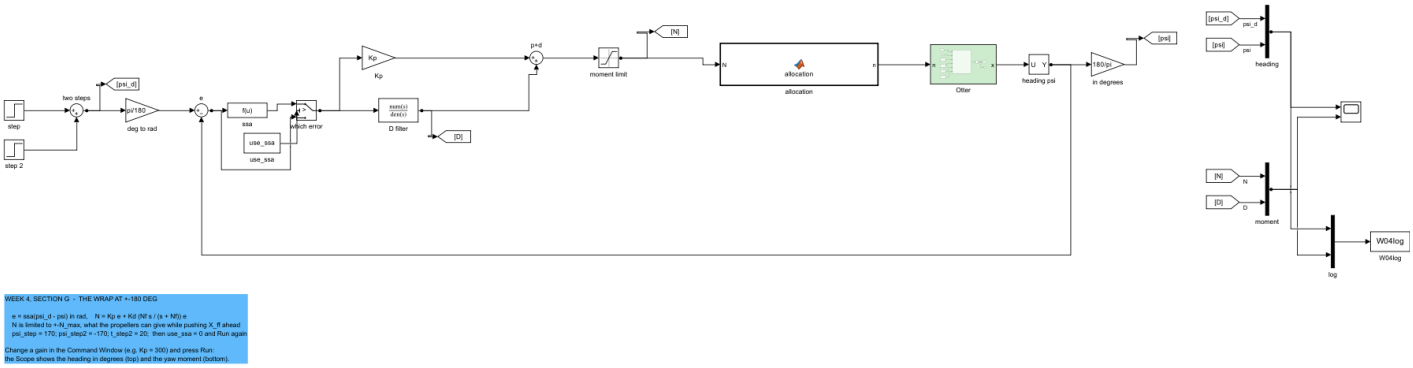
Experiment 4-6 · The wrap (10 min)

What it measures. Lines 2 and 3: the turn the vessel actually performs with the wrap in the error path and with it removed, for one and the same command.

The model. `W04_G_wrap` — the tuned loop with a switch, `use_ssa`, between the wrapped error and the raw difference. Nothing else differs between the two runs.

Opening and running.

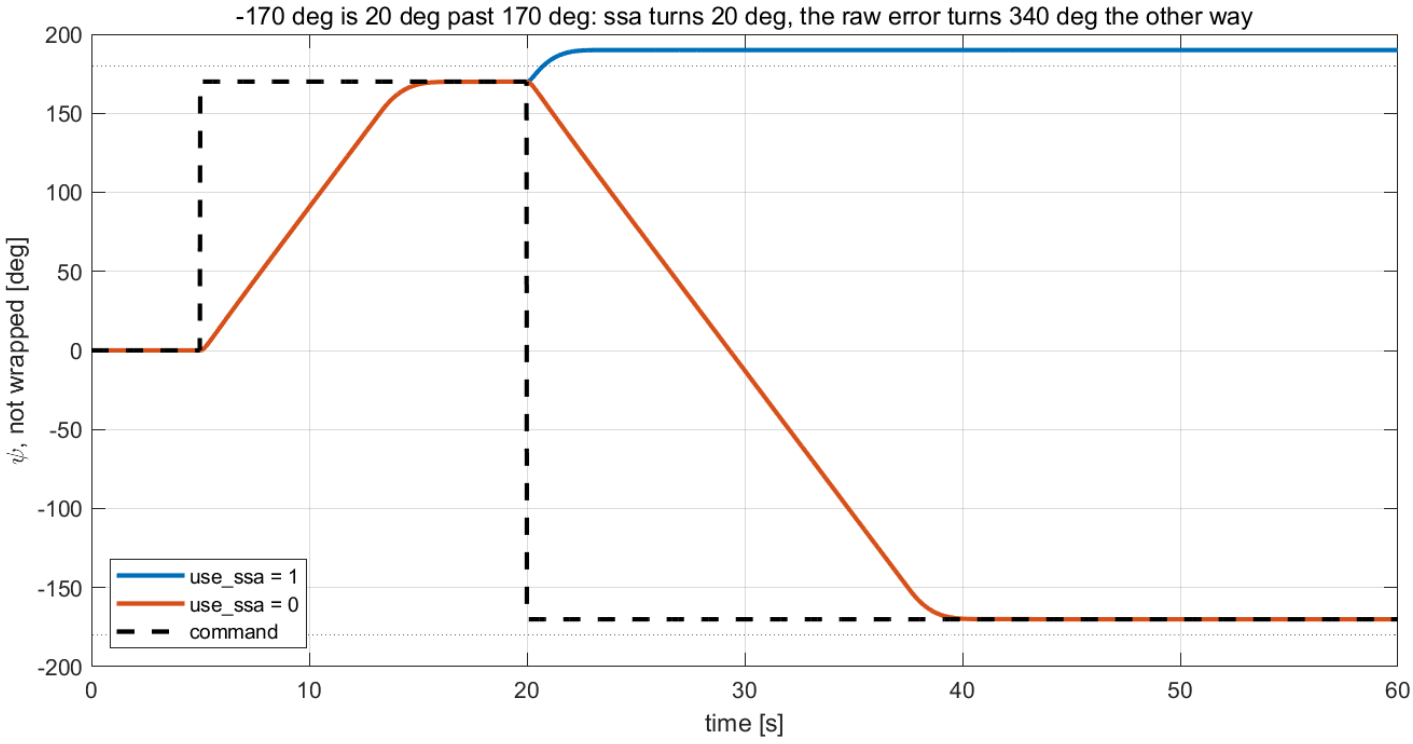
```
W04_0_setup
psi_step = 170; psi_step2 = -170; t_step2 = 20; T_final = 60;
open_system('W04_G_wrap')          % Run — 20 도만 돌린다 / a 20 deg turn
use_ssa = 0;                       % Run — 반대로 340 도 / 340 deg, the other way
W04_G_the_wrap                     % 두 실행을 한 번에 / both runs at once
```



In the figure	Meaning
`ssa` , `use_ssa` , `which error`	the wrapped error when `use_ssa = 1` , the raw error when 0
`step` , `step 2` , `two steps`	170° at 5 s, then -170° at `t_step2`

Expected output:

```
W04 Experiment 4-6  170 deg, then -170 deg at t = 20 s
  use_ssa   heading at 20 s [deg]   heading at 60 s [deg]   turned after 20 s [deg]
  1                         170.0                   190.0                     20.0
  0                         170.0                  -170.0                    340.0
```



In the figure	Meaning
traces	the heading, not wrapped, with and without `ssa` ; the dotted lines are $\pm 180^\circ$

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
both traces **up to** $t = 20$ s	identical, at 170°	the seam has not been crossed yet, and the two error paths agree everywhere else
the two traces **leaving** 170° in opposite directions	$+20^\circ$ and -340°	line 2: the controller turns the way the number it was handed points
where each trace **ends**	190° and -170°	line 1: the same heading, written two ways. The plot leaves the heading unwrapped so that the turn is visible
the 340° trace **crossing** the dotted $\pm 180^\circ$ lines	it sweeps the whole circle	line 3: what `ssa` exists to prevent

What the figure says

- 190° and -170° are the same heading. With `ssa` the vessel reaches it by a 20° turn; without it, by 340° the other way.

What to try	What to watch
`psi_step2 = 150;` with `use_ssa = 0`	no seam is crossed, and the two settings agree exactly: the bug hides until the command happens to cross $\pm 180^\circ$
`psi_step = -170; psi_step2 = 170;` with `use_ssa = 0`	the same fault in the other direction — a 340° turn to starboard

> ✎ **In class**
>
> - **Purpose** — one line of code, and the cost of leaving it out.
> - **Point to** — the two curves leaving 170° in opposite directions at 20 s.
> - **Ask** — "Why does the plotted heading end at 190° and not at -170°?" The plot shows the heading without wrapping, to make the direction of the turn visible; physically they are the same.
> - **Take away** — wrap every heading error before any gain sees it.

4-7. The inner loop, assembled: two errors, two forces, two propellers

What is observed. Four sections have each fixed one piece of the heading law, and they have been fixed in different places: the wrap in §4-6, the derivative in §4-3d, the limit in §4-0. Written on one line the result is short enough to read aloud, and it is also the complete contents of the `heading autopilot` block:

$$\boxed{N = K_p\,\mathrm{ssa}(\psi_d - \psi) \;-\; K_d\,r}, \qquad |N| \le N_{\max}$$

Symbol	Quantity	Value · source
ψ_d	commanded heading	rad; a step this week, a guidance law in Week 5
ψ	heading, state 12	rad
r	yaw rate, state 6 — **measured, not differentiated**	rad/s
K_p	proportional gain	300 N·m per rad, `W04_0_setup`
K_d	rate gain	100 N·m per rad/s, `W04_0_setup`
$N_{\max}$	moment limit, from what the propellers can give	70.85 N·m, §4-0

The result. Three decisions are visible in that one line, and each was forced by a measurement earlier in the week:

In the law	The decision	Where it was forced
$\mathrm{ssa}(\cdot)$ around the error	wrap into $(-\pi, \pi]$ **before any gain sees it**	§4-6: without it a 20° turn becomes a 340° turn
$-K_d\, r$, not $+K_d\, \dot{e}$	take the derivative from the **measurement**, not from the error	§4-3d: the filtered error form kicks by $K_d N_f$ on a step
$\lvert N \rvert \le N_{\max}$	clamp to what two propellers can actually produce	§4-0, and §3-4's windup is what happens when it is ignored

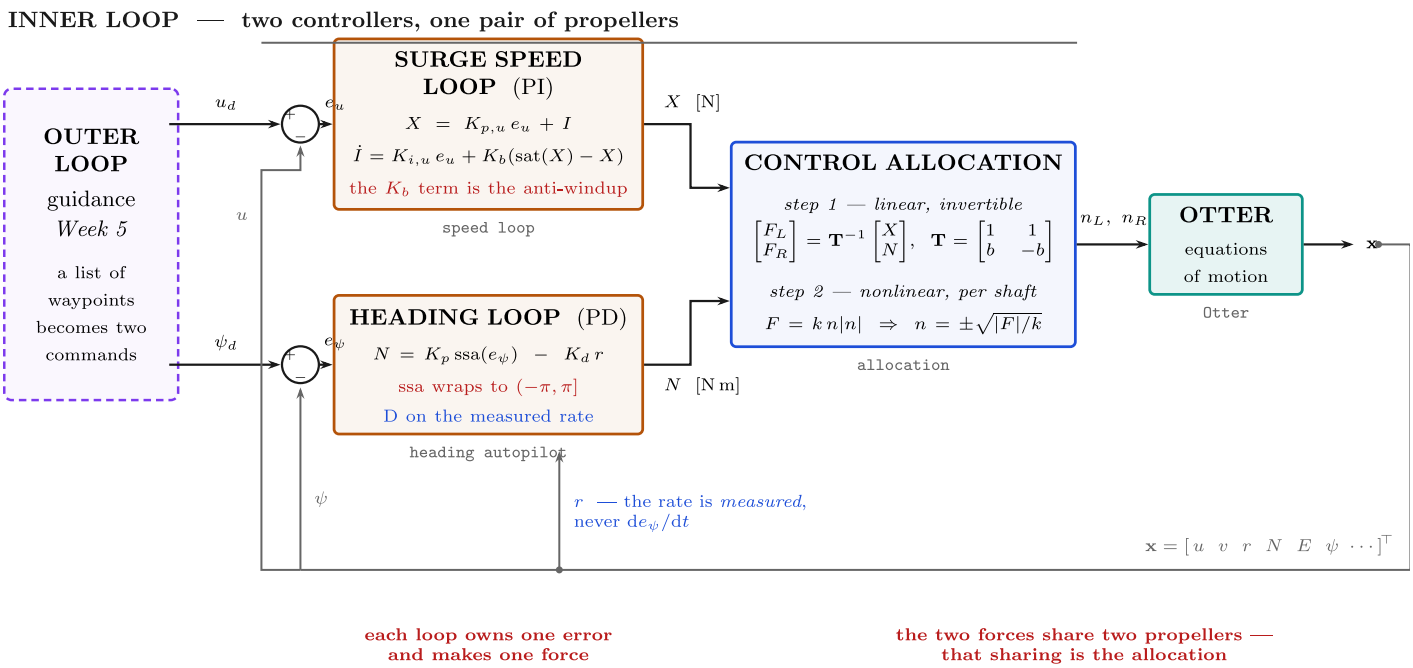
Derivation of the sign, in one line. The D term is often written $+K_d\, \mathrm{d}e_\psi/\mathrm{d}t$. With $e_\psi = \psi_d - \psi$ and a constant command, $\dot{e}_\psi = -\dot{\psi} = -r$, so

$$+K_d \frac{\mathrm{d}e_\psi}{\mathrm{d}t} \;=\; -K_d\, r$$

The two forms are **the same term**, and the minus sign is not a convention to memorise: it is what the chain rule returns. What differs is only what happens when the command is *not* constant, and that difference is the whole of §4-3d — the error form passes $\dot{\psi}_d$ into the moment and the rate form does not.

> **★ How big the kick actually is**
>
> On a unit step of ψ_d the filtered-error form asks for $K_d N_f = 100 \times 20 = 2000$ N·m at the instant of the step. The moment limit is $N_{\max} = 70.85$ N·m. The demand is therefore **28.2 times everything the vessel has**, all of it thrown away by the saturation, and none of it steering. That number is why this course writes the rate form.

Where that moment goes. The output of this block is a number in newton-metres, and the Otter has no actuator that accepts newton-metres. §3-6 built the two steps that turn it into shaft speeds, and this week supplies the input that section left at zero:

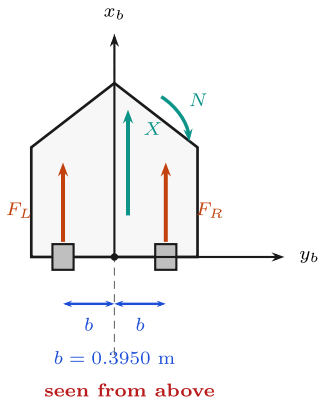


In the diagram	This week's part
the lower row, `heading autopilot`	the law boxed above: $\mathbf{ssa}$ on the error, K_d on the measured rate
the feedback marked r	goes **straight into the D term**, not through the summing junction — the diagram shows the two paths separately for exactly this reason
N entering `allocation`	the input §3-6 set to zero; it is no longer zero
the upper row	Week 3, unchanged and not retuned

With $N \neq 0$ the matrix of §3-6 line 2 finally does something:

$$F_{L,R} = \frac{X}{2} \pm \frac{N}{2b}$$

The surge force is still shared equally; the moment is **added to the port propeller and taken from the starboard one**, which is what makes the vessel turn to starboard for $N > 0$. Where that matrix comes from is §3-6, and this is the figure to re-read now that N is no longer zero:



1. read the three forces off the picture

$$\begin{bmatrix} X \\ Y \\ N \end{bmatrix} = \begin{bmatrix} F_L + F_R \\ 0 \\ b\,(F_L - F_R) \end{bmatrix}$$

both push forward, so they **add**

neither pushes sideways:
this row is empty, always

opposite arms, so they **subtract**

2. drop the empty row — it is square now

$$\begin{bmatrix} X \\ N \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ b & -b \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} F_L \\ F_R \end{bmatrix}$$

$\mathbf{T}$ is **geometry**, nothing else:
it never changes while the
propellers stay bolted where they are

3. invert it — that is the allocator

$$\begin{bmatrix} F_L \\ F_R \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} X \\ N \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2b} \\ \frac{1}{2} & -\frac{1}{2b} \end{bmatrix} \begin{bmatrix} X \\ N \end{bmatrix}$$

known

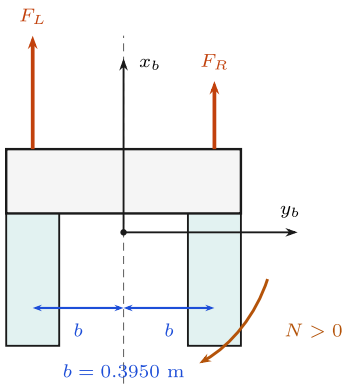
what the two
controllers asked for

$\det \mathbf{T} = -2b \neq 0$, **so the inverse exists and the split is unique: each propeller takes half of X, and the moment is added to one and taken from the other**

In the figure	What it means now that $N \neq 0$
the teal N arrow on the hull	this week's output, and the first week it is not zero
the third row, $b\,(F_L - F_R)$	the moment comes from the **difference**, so commanding one means driving the two propellers apart
the empty middle row	still empty: the hull makes no sideways force, so a heading is the **only** way it goes anywhere but straight — which is what Week 5 will build on
line 3, $\frac{1}{2b}$ and $-\frac{1}{2b}$	the moment divided by the lever, added to one propeller and taken from the other. A short lever costs thrust: $N = 70.85$ N·m over $2b = 0.79$ m needs 89.68 N of difference

The second step, the propeller curve, is where a demanded thrust becomes a shaft speed, and it is the step that decides whether the demand was possible at all:

STEP 1 two thrusts into two forces, and back



where the two thrusts act

both push forward, and the two arms are opposite

$$\begin{bmatrix} X \\ N \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ b & -b \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} F_L \\ F_R \end{bmatrix}$$

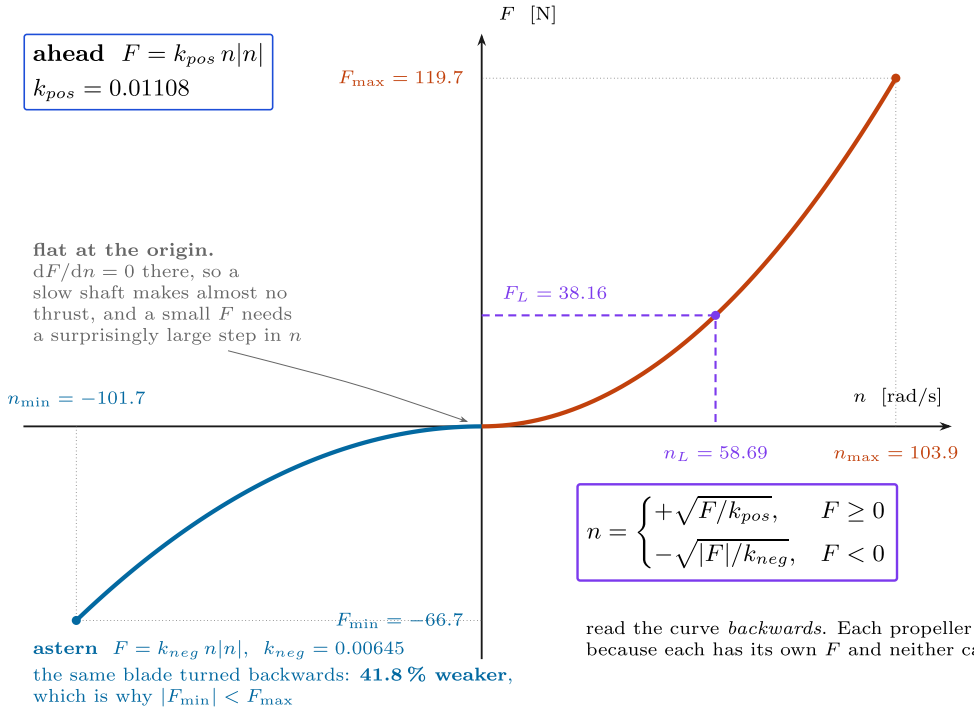
$\det \mathbf{T} = -2b \neq 0$: the matrix is invertible, so the split is *unique*

$$\begin{bmatrix} F_L \\ F_R \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} X \\ N \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2b} \\ \frac{1}{2} & -\frac{1}{2b} \end{bmatrix} \begin{bmatrix} X \\ N \end{bmatrix}$$

each propeller carries *half* the surge force, and the moment is added to one and taken from the other

linear, and exactly invertible

STEP 2 each force onto its own shaft — nonlinear, one propeller at a time



ahead $F = k_{pos}\, n|n|$
$k_{pos} = 0.01108$

flat at the origin.
$\mathrm{d}F/\mathrm{d}n = 0$ there, so a slow shaft makes almost no thrust, and a small F needs a surprisingly large step in n

astern $F = k_{neg}\, n|n|$, $k_{neg} = 0.00645$
the same blade turned backwards: **41.8 % weaker**, which is why $|F_{\min}| < F_{\max}$

$$n = \begin{cases} +\sqrt{F/k_{pos}}, & F \geq 0 \\ -\sqrt{|F|/k_{neg}}, & F < 0 \end{cases}$$

read the curve *backwards*. Each propeller is read on its own, because each has its own F and neither can help the other

nonlinear, but still one-to-one: every reachable F has exactly one n

In the figure	What this week adds
STEP 1, top	the matrix of §3-6, now with a non-zero N in it
STEP 2, the orange branch	where the **loaded** propeller sits during a hard turn — at the very top of it
STEP 2, the blue branch	where the **unloaded** one goes: astern, on the weaker curve
the marked points	$F_{\max} = 119.7$ N and $F_{\min} = -66.7$ N, the two limits that $N_{\max}$ was solved against

Putting this week's numbers through both steps, at the hardest point of a 30° turn:

Quantity	Value	Where from
X	60.00 N	the constant `X_ff` of §4-0
N	70.85 N·m	the clamp, saturated
$F_L = X/2 + N/(2b)$	$30.00 + 89.68 = 119.68$ N	$= F_{\max}$, exactly
$F_R = X/2 - N/(2b)$	$30.00 - 89.68 = -59.68$ N	astern, and inside $F_{\min} = -66.7$ N
$n_L = \sqrt{F_L/k_{pos}}$	103.93 rad/s	$= n_{\max}$, exactly
$n_R = -\sqrt{\lvert F_R\rvert/k_{neg}}$	-96.23 rad/s	inside $n_{\min} = -101.7$

- $N_{\max}$ was not chosen; it was **computed** so that the loaded propeller lands exactly on its limit and not past it. That is the calculation in `W04_0_setup`, and §4-0's table is where its value came from.

The same two steps, inside the model

Both steps live in one block. Below is the code it holds, copied out of `W04_H_tuning.slx`; double-clicking `allocation` in any model of this week shows the same text. The comments are the block's own.

```
function n = allocation(N, X_ff, y_pont, k_pos, k_neg, port_eff)
%#codegen
%ALLOCATION  요 모멘트 N [N m] 를 두 프로펠러의 축 회전수 n [rad/s] 로 바꾼다.
%            Turn the yaw moment N [N m] into the shaft speeds n [rad/s] of the two propellers.
%
%   N 만 입력 포트다. 나머지는 작업공간 변수 (W04_0_setup) 에서 온다.
%   Only N is an input port; the rest come from the workspace (W04_0_setup).
%     X_ff      두 프로펠러가 함께 내는 전진력 / surge force from both propellers   [N]
%     y_pont    중심선에서 각 프로펠러까지의 거리 / each propeller's arm              [m]
%     k_pos     앞으로 밀 때 추력 계수 / thrust coefficient ahead
%     k_neg     뒤로 당길 때 추력 계수 / thrust coefficient astern
%     port_eff  좌현 프로펠러 효율, 보통 1 (절 F 는 0.7) / port efficiency, 1 unless section F

% 1) 두 프로펠러의 추력. 둘 다 X_ff 의 절반씩 밀고, 모멘트는 한쪽을 더 세게,
%    다른 쪽을 덜 세게 밀어 만든다. 팔 길이가 y_pont 이므로
%        y_pont (T1 - T2) = N,   T1 + T2 = X_ff
%    Both push half of X_ff ahead; the moment comes from pushing one side
%    harder and the other less. With arm y_pont:
%        y_pont (T1 - T2) = N,   T1 + T2 = X_ff
T = [X_ff/2 + N/(2*y_pont);      % 좌현 / port
     X_ff/2 - N/(2*y_pont)];     % 우현 / starboard

% 2) 약한 좌현 프로펠러: 요구한 추력의 port_eff 배만 낸다 (절 F).
%    A weak port propeller delivers only port_eff of what is asked (section F).
T(1) = port_eff * T(1);

% 3) 프로펠러 곡선 T = k n|n| (1주차) 을 하나씩 n 에 대해 푼다.
%    앞으로와 뒤로의 계수가 다르다 (k_pos > k_neg).
%    Solve the propeller curve T = k n|n| (Week 1) for each n; the coefficient
%    differs ahead and astern (k_pos > k_neg).
n = zeros(2,1);
for i = 1:2
    if T(i) >= 0
        n(i) =  sqrt( T(i) / k_pos);   % 앞으로 / ahead
    else
        n(i) = -sqrt(-T(i) / k_neg);   % 뒤로 / astern
    end
end
end
```

In the code	In the figure	Note
the two rows of `T = [...]`	the row $\left[\frac{1}{2} \quad \frac{1}{2b}\right]$ and $\left[\frac{1}{2} \quad -\frac{1}{2b}\right]$ of $\mathbf{T}^{-1}$, STEP 1	$\mathbf{T}^{-1}$ written out by hand, with $b =$ `y_pont`. §5-7 writes $\mathbf{B}$ down instead and lets MATLAB solve it; the arithmetic is the same
`N` as the only input port	the teal N arrow	the P–D law of §4-3 produces it; `X_ff` is a constant this week, and Week 5 replaces it with the speed loop's output
`T(1) = port_eff * T(1);`	nothing — the figure draws the healthy vessel	§4-F's damaged propeller. With `port_eff = 1` the line does nothing, which is why it can sit in the shared block
the `for i = 1:2` loop	both branches of STEP 2, once per propeller	each propeller is read off the curve **on its own**: neither can help the other, and that is what makes the limit per-propeller rather than per-vessel
`T(i) >= 0` versus `else`	the orange and blue branches	at the hardest point of the turn the loop takes the orange branch for the port propeller and the blue one for the starboard, in the same time step

To check every number in this section

```
cd _tools
verify_w04_heading
```

Actual output:

```
verify_w04_heading

  1 W04_vars = W04_0_setup                  OK
  2 N_max = 70.85 N m                       OK    70.8488
  3 thrust split returns X_ff and N         OK    7.1e-15
  4 atan2(sin, cos) = MSS ssa               OK    4.9e-16
  5 PD leaves no heading error              OK    0.0009 deg

ALL CHECKS PASSED
```

The check	What it proves
2	the moment limit is the one the propeller limits imply, not a round number someone picked
3	feeding (F_L, F_R) back through $\mathbf{T}$ returns the (X, N) they came from, to 7×10^{-15} — the inverse of §3-6 line 2 is exact
4	the `ssa` block agrees with the MSS function of the same name to 5×10^{-16}
5	a P–D law with no integral still leaves no steady heading error, because yaw has no restoring moment to fight (§4-3)

What to try	What to watch
`Kd = 0;` then Run `W04_F_PID`	the rate term removed: the heading rings, and the overshoot is the measurement §4-3b made
`Kp = 600;` Run	N reaches its clamp sooner and stays there longer; the turn does not get proportionally faster, because $N_{\max}$ has not moved
In `W04_F_PID`, double-click `heading autopilot`	the boxed law, block by block: one `Fcn` for `ssa`, a `Gain` for each term, and a `Saturation` holding `N_max`

> ✎ **In class**
>
> - **Purpose** — put the week's four separate decisions into one line and watch that line reach the propellers.
> - **Point to** — the r arrow in the diagram, entering the controller from below rather than through the summing junction.
> - **Ask** — "Why is F_L exactly 119.68 N at the clamp?" Because $N_{\max}$ was solved for that condition: the limit is the largest moment that does not ask a propeller for more than it has.
> - **Ask** — "The starboard propeller is running astern at -96 rad/s while the vessel moves forwards. Is that a mistake?" No — that is how a differential-thrust vessel turns hard, and it is also why a hard turn costs speed.
> - **Take away** — the controller's output is a force, not a command to an actuator; everything between the two is arithmetic with limits in it.

4-8. Both inner loops at once: the speed loop joins the heading loop

What is observed. Every model so far pushed the hull with a constant $X_{ff} = 60$ N. The heading was controlled and the speed was not: 0.77 m/s was never commanded by anyone, it is simply where thrust and drag balance. Putting the Week 3 speed loop in that place gives the vessel a second controller, and the two then draw on the same two propellers.

The result. Both loops hold their commands, and they do so through each other's transients. The price is paid in the limits: the moment clamp must be recomputed, because the surge force is no longer a known constant.

Derivation.

1. **Two laws, unchanged.** Neither controller is modified. The heading law is §4-7's, and the speed law is Week 3's, integrator and anti-windup included:

$$N = K_p\ \mathrm{ssa}(\psi_d - \psi) - K_d\, r, \qquad X = K_{p,u}(u_d - u) + I, \qquad \dot{I} = K_{i,u}(u_d - u) + K_{b,u}\big(X - X_{raw}\big)$$

Symbol	Quantity	Value · source
K_p, K_d	heading gains	300, 100; `W04_0_setup`, §4-5
$K_{p,u}, K_{i,u}$	speed gains	200, 200; the Week 3 values
$K_{b,u}$	the speed loop's back-calculation gain	1 s^{-1} — **not** the `Kb` of §4-5, which belongs to the heading integrator
$X_{\max}$	the surge force limit	120 N, both propellers together

2. **One allocator, two live inputs.** §4-7's allocator already takes X and N; what changes is that both now arrive as signals rather than one of them being a constant:

$$\begin{bmatrix} T_L \\ T_R \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} X \\ N \end{bmatrix}, \qquad n_i = \pm\sqrt{|T_i|/k}$$

3. **The clamp has to move.** $N_{\max} = 70.85$ N·m was computed at $X = X_{ff} = 60$ N. With a speed loop that can ask for $X_{\max} = 120$ N, the moment that remains available is smaller, and the `Saturation` in the heading loop must hold that smaller number instead:

$$N_{speed} = \min\left\{2y_p\left(T_{\max} - \tfrac{X_{\max}}{2}\right),\ 2y_p\left(\tfrac{X_{\max}}{2} - T_{\min}\right)\right\} = 47.15\ \text{N·m}$$

Leaving $N_{\max}$ in place would let the allocator ask for a shaft speed the propeller does not have. Week 5 §5-7 derives this expression in full and measures what it costs; this section is where the rule first applies.

4. **Why it is here and not in Week 5.** The guidance of Week 5 produces **two commands at once** — a heading and a speed. The pair of loops that receives them is this model. Checking the inner pair before any guidance is attached means that when the outer loop misbehaves next week, the inner one is already known to be innocent.

Experiment 4-8 · Two commands, two loops, one hull (10 min)

Both commands are staircases from `W04_0_setup`, each value held for 30 s, so what the figure shows is tracking rather than a single step response:

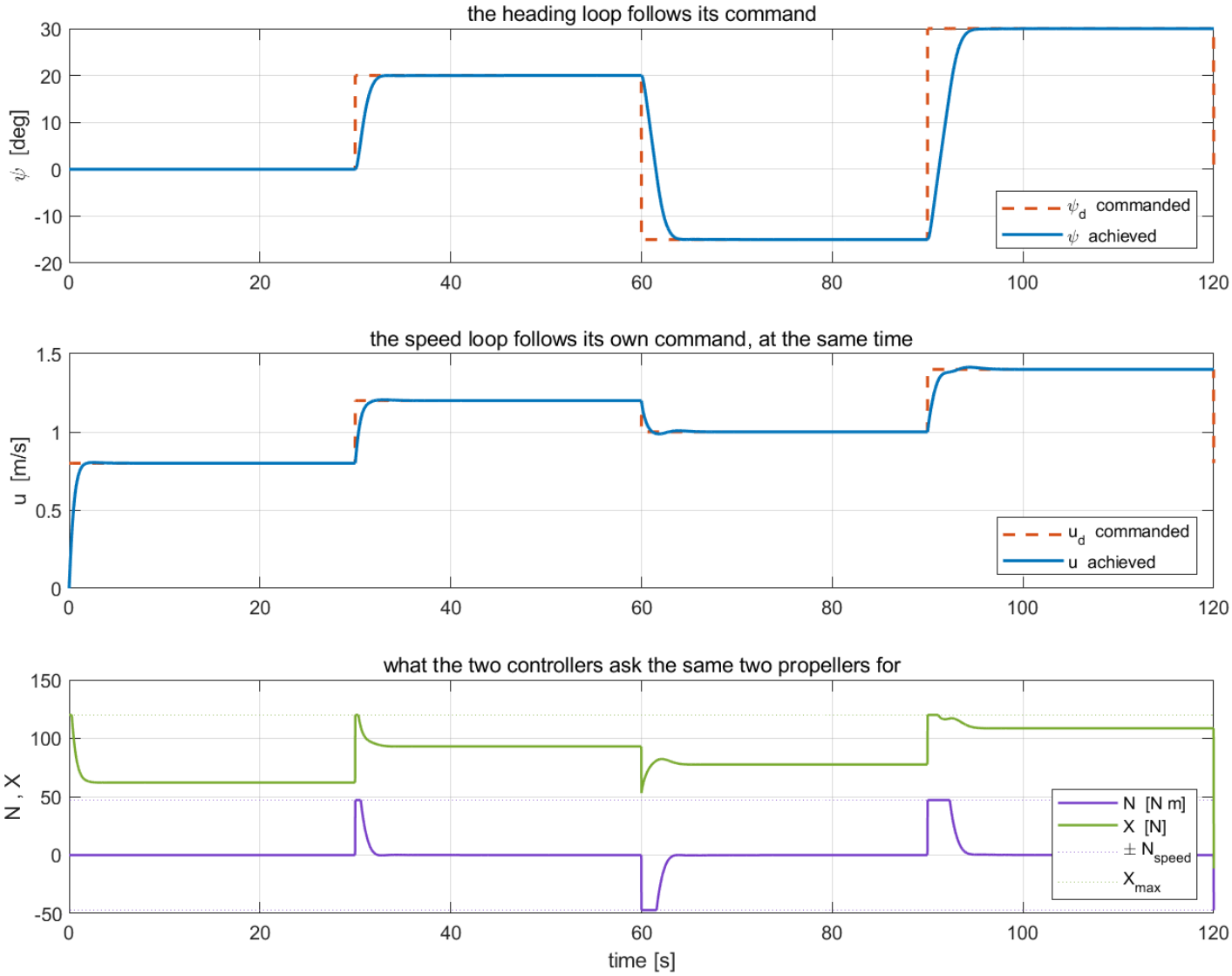
Command	Values	Held for
u_d	$0.8 \to 1.2 \to 1.0 \to 1.4$ m/s	30 s each
ψ_d	$0 \to 20 \to -15 \to 30$ deg	30 s each

```
cd lectures/W04_simulink
W04_0_setup
W04_J_build_both          % once, to write the model
W04_J_speed_and_heading
```

Actual output:

```
W04 Experiment 4-8  both inner loops at once  (Kp = 300, Kd = 100, Kp_u = 200, Ki_u = 200)
  step   psi_d [deg]   psi end [deg]   u_d [m/s]   u end [m/s]   max |N| [N m]   max X [N]
   1           0.0           -0.00          0.80         0.800            0.00         120.0
   2          20.0           20.00          1.20         1.200           47.15         120.0
   3         -15.0          -15.00          1.00         1.000           47.15          82.3
   4          30.0           30.00          1.40         1.400           47.15         120.0

  the two limits, and how long each was reached
    |N| at N_speed = 47.15 N m  for  4.54 s
    X   at X_max   = 120.0 N    for  1.74 s
```



In the figure	Meaning
panel 1	the commanded heading and the one achieved, in degrees
panel 2	the commanded speed and the one achieved, at the same instants
panel 3	N and X, each against its own limit

Reading the figure against the derivation.

Where to look	What is there	Which line it settles
`psi end` column	20.00, -15.00, 30.00 deg against commands of the same	line 1: no steady error, with no integrator in the heading law — §4-3's point, now with the speed changing underneath it
`u end` column	0.800, 1.200, 1.000, 1.400 m/s	line 1: the speed loop's integrator is what makes these exact; drag would otherwise leave an offset
panel 1 at $t = 60$ s	the heading swings 35° while the speed command also changes	the two commands move at the same instant and neither loop is disturbed by the other
panel 2, the dip after each corner	a few hundredths of m/s, recovered within seconds	a hard turn costs speed, and the integrator buys it back
panel 3, N at every corner	flat at ± 47.15 N·m	line 3: the clamp is `N_speed`, not `N_max`
panel 3, X at the same moments	at 120 N, its own limit, for 1.74 s in all	both controllers reach their limits in the same seconds; that is what sharing two propellers means
step 3's row	$\max X = 82.3$ N, below the limit	the only step that asks for **less** speed; the force needed is smaller and nothing saturates

What the figure says

- Two controllers on one hull do not interfere while both stay inside their limits — and the limits are where they meet, which is why the moment clamp had to be recomputed before the speed loop was allowed to exist.

What to try	What to watch
In `W04_0_setup` set `N_speed = N_max;` then rerun	the allocator asks for more than the starboard propeller has at every corner; `n` is clamped and the heading response stops matching the law
`u_d_steps = [1.4 1.4 1.4 1.4];`	the speed loop holds one value and the corners alone disturb it — the dips of panel 2 without the steps
`Kb_u = 0;`	the back-calculation removed: the speed overshoots after each step that saturates the force
Double-click `speed loop`, then `heading autopilot`	the two subsystems are the diagrams of Weeks 3 and 4 unchanged — both are built by `_tools/add_speed_loop` and `_tools/add_heading_loop`, which Week 5 calls as well

> ✎ **In class**
>
> - **Purpose** — show that the two inner loops are independent in operation and coupled only through the actuator limits.
> - **Point to** — panel 3 at $t = 90$ s: both traces are flat against their limits at the same moment.
> - **Ask** — "Which loop wins when both want more?" Neither: the allocator splits what is asked, and each loop's own Saturation is what keeps the demand feasible.
> - **Take away** — the pair of commands that Week 5's guidance will produce already has somewhere to go.

Part 2 · Laboratory run order

The experiments of Part 1 are worked through in order; this table is the index of what was run, for repeating the week at home.

Every example has **its own model**, laid out as in Weeks 2 and 3: the controller on the left, the thrust split and the vessel in the middle, **one Scope** on the right with the heading in degrees on top and the yaw moment below.

Experiment	Model	Script	What it shows
4-0	all of them	`W04_0_setup`, `W04_1_build_heading`	the parameters, and every model written from code
4-1	`W04_H_tuning`	—	the Week 3 canvas with `ssa` in front and the thrust split behind
4-2	`W04_C_open_loop`	`W04_C_plant_from_outside`	a step yaw moment, no controller
4-3a	`W04_D_P`	`W04_D_proportional_only`	P only
4-3b	`W04_E_PD`	`W04_E_derivative`	P + D
4-3c	`W04_F_PID`	`W04_F_weak_propeller`	P + I + D, with a weak port propeller
4-3d	`W04_E_PD`, reread	—	both D forms read out of one run: the kick, and what the rate form would have asked
4-3e	`W04_I_rate`	`W04_I_rate_feedback`	the three D terms, each one actually steering: the gyro, the error, and the wrong sign
4-4	none — two lines in the Command Window	—	the crab angle from u and v
4-5	`W04_H_tuning`	`W04_H_tuning_by_hand`	the tuning order, a big turn, and the back-calculation gain
4-6	`W04_G_wrap`	`W04_G_the_wrap`	a command across $\pm 180^\circ$, with an `ssa` switch
4-7	`W04_F_PID`	`verify_w04_heading`	the assembled law, and the moment followed to the two shafts
4-8	`W04_J_both`	`W04_J_build_both`, `W04_J_speed_and_heading`	the speed loop and the heading loop together, each following a staircase command

- The scripts only repeat what Run already shows, for several gains at once, and print the numbers quoted in Part 1.
- Each experiment ends with an **In class** note: purpose, what to point at, a question with its answer, and the sentence to take away.

Summary

Week Summary

Step	What was done	How it was verified
1	placed the Week 2 controller around the heading, with `ssa` and a thrust split	`check_overlaps` 0 in seven models
2	measured the yaw axis from outside	Experiment 4-2: the heading grows without end; turn rate within 0.36 s to 63 %; rate per N·m falls from 0.610 to 0.407
3	P alone	Experiment 4-3a: no error at any gain; overshoot $5.8 \to 15.9\,\%$
4	D damps	Experiment 4-3b: $12.18 \to 0.39\,\%$, fastest at $K_d = 100$
5	I against a weak propeller	Experiment 4-3c: 0.80° removed; integral 4.15 N·m
5a	settled where the D term comes from, and why its sign is negative	Experiment 4-3e: the rate and error forms track alike (0.90 against $0.39\,\%$) and demand 17.03 against 325.79 N·m; the sign flipped never settles, limit-cycling at 7.23 deg/s against the 7.70 predicted
6	the wrap	Experiment 4-6: 20° with `ssa`, 340° without
7	the tuning order and K_b	Experiment 4-5: $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$; 90° turn in 5.80 s without overshoot

Progress Check

> ★ **Minimum condition for following Week 5**

Theory

- [ ] Able to explain from one open-loop test why P alone leaves no heading error.
- [ ] Able to state what P, I and D do on the heading and compare with Weeks 2 and 3.
- [ ] Able to compute the crab angle and course from u, v and ψ.
- [ ] Able to explain both failure modes of the back-calculation gain.

Laboratory

- [ ] `W04_1_build_heading` ran and every model reported `overlapping lines: 0`.
- [ ] `W04_check(1)`, `(2)` and `(3)` pass on a model built from `W04_P1_start`.

Recorded observations

- [ ] The heading error left by PD with a 70 % port propeller, and the moment the integral found.
- [ ] The 90°-turn settling time at $K_b = 0$, 0.1 and 1.

In-class laboratory — build the heading autopilot by hand

The second hour of the Week 4 session is spent building the autopilot in Simulink around a vessel whose allocation is given.

```
cd lectures/W04_simulink/problems
W04_P1_start                % creates W04_P1.slx — hull and allocation only
W04_check(1)                % run this whenever, as often as needed
```

	Problem	Time	The number it must reproduce
1	P only	20 min	no error; overshoot $5.8\,\%$ at $K_p = 100$, $12.2\,\%$ at 300
2	add the filtered derivative	20 min	overshoot 12.18, 4.10, $0.39\,\%$ at $K_d = 0$, 50, 100
3	the wrap	20 min	a 20° turn with `ssa`, 340° without

- The problem sheet is `W04_simulink/problems/README.md`, and reference answers are in `W04_simulink/solutions/`.

Assignment 4

- **Due**: before the Week 5 session
- **Submit**: the modified `W04_0_setup.m`, the numbers requested below, and two figures

① Requirements

Halve the surge force, `X_ff = 30` N, in `W04_0_setup.m` and `W04_vars.m`, and tune the heading loop again by the tuning order of §4-5 with the same requirement.

② Verification — mandatory

> **★ A claim is not a result. Numbers are required, with the tolerance band stated**

1. The new $N_{\max}$, computed from §4-1 and read from `W04_0_setup`'s output.
2. Section C with the new X_{ff}: the turn rate per N·m at 10 N·m.
3. For each step of the tuning order, the measurement that decided it.
4. The K_b table of §4-5 for the final gains.

③ Analysis (5–10 lines)

Explain why the available yaw moment depends on the surge force, and how that moved the gains.

Grading

Criterion	Weight
The model runs and produces the requested output	25%
Verification performed and numbers reported	40%
Correctness of the analysis	25%
Readability of the code	10%

Troubleshooting

Symptom	Cause	Fix
`W04_0_setup` stops in `mss_path`	the MSS toolbox is not at `Tools\MSS`	place MSS there; see the first page
the vessel stalls short of a big turn for tens of seconds	K_b too large	`Kb = 0.1`
a big turn overshoots and creeps back	$K_b = 0$: windup	`Kb = 0.1`
the vessel turns the long way round	`use_ssa = 0`	`use_ssa = 1`
a Scope does not match these notes	a changed variable is still in the workspace	run `W04_0_setup` again
`allocation` reports that `X_ff` or `k_pos` is undefined	`W04_0_setup` was not run, so the block's parameters have no values	run `W04_0_setup`; the block reads `X_ff`, `y_pont`, `k_pos`, `k_neg` and `port_eff` from the workspace

---

References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley, 2021, §12.2 (PID control of marine craft) and §12.2.6 (anti-windup); the crab angle and course are defined with the kinematics of Ch. 2.
- Åström, K. J. and Hägglund, T. *Advanced PID Control*, ISA, 2006, Ch. 3.
- MSS toolbox, `Tools/MSS/VESSELS/otter.m` and `GNC/ssa.m`. The rate form of the D term is what the toolbox itself steers by — `VESSELS/SIMclarke83.m`, `VESSELS/SIMremus100.m` and `VESSELS/SIMrig.m`; not one MSS vessel differentiates the heading error.

In this course

- Week 2 §2-4, §2-6 to §2-8, §2-11 and §2-12; Week 3 §3-3 and §3-5.
- The previous, model-based version of this week — the Nomoto model, pole placement and the second-order formulas — is kept in `W04_simulink/_previous_version/` for reference.

Next Week

- **Week 5 — Waypoint Following and LOS Guidance**
- A heading autopilot follows a commanded heading; guidance decides what heading to command. Week 5 builds the line-of-sight law that turns a list of waypoints into ψ_d, and meets the crab angle of §4-4 as a cross-track error.